of the therapy increases, the likelihood of muscle fatigue also increases. A lower frequency (35–50 Hz) will provide strong muscle contractions while reducing muscle fatigue and discomfort (Windsor *et al.*, 1993; Nelson *et al.*, 1999; Levine and Bockstahler, 2014). An alternative with veterinary patients that are not very tolerant of electrical stimulation would be to use a low frequency (<10 Hz), which will provide only a twitch (localized) contraction, but will possibly be more easily tolerated while still stimulating some function.

On/Off or Duty Cycle

The duty cycle is stated in seconds or in a ratio form of "on-time" and "off-time." Common clinical parameters for on/off times are 1:3, or 10 seconds on:30 seconds off, but can be modified to accommodate the patient's needs and goals of treatment (Bracciano, 2008; Levine and Bockstahler, 2014). The duty cycle settings can be used to decrease the likelihood of premature muscle fatigue during a therapy session. The "on-time" is when the series of pulses are being delivered. The "off-time" is the time between sequential on-times and during this no current is being delivered. The off-time phase allows the muscle tissues to recover (relax and reset for next contraction), adequate recovery time decreases the chance of muscle fatigue. During this reset period, the adenosine triphosphate (ATP)/adenosine diphosphate (ADP) are replenished. Intermittent electrical simulation is commonly used to increase the comfort level of the patient (Lake, 1992; Doucet *et al.*, 2012). On–off cycles are not a part of continuous flow electrical stimulation (Box 16.2).

Ramp

This is the gradual increase or decrease of time from when the stimulus reaches peak amplitude and is usually measured in seconds (Kloth and Cummings, 1991). Off-time is the time between on-times. At the onset of the electric current, a gradual increase in flow is designed to slowly increase the force of muscle contraction by gradually recruiting motor units. A steady decrease in current at the end of the peak amplitude results in a smooth decline in muscle force. A 1–3 second ramp time is commonly utilized in clinical applications and the amount of time is based on patient comfort. If a patient has increased muscle tone that may create resistance against the stimulated movement, then a longer ramp is utilized (Bracciano, 2008). In some situations, using a ramp time may

Box 16.2 Typical parameters available in neuromuscular electrical stimulation devices

- *Waveform*: The shape of the visual representation of pulsed current on a current/time plot or voltage/time plot. Can be symmetric, asymmetric, balanced, unbalanced, biphasic, monophasic, or polyphasic
- *Amplitude*: The current value in a monophasic pulse or for any single phase of a biphasic pulse
- *Phase/pulse duration*: The duration of a phase or a pulse, usually measured in microseconds
- *Pulse rate or frequency*: The rate of oscillation in cycles per second, expressed as pulses per second (pps) or hertz (Hz). Often labeled as pulse rate or pulses per second, or frequency on stimulators
- *On/off time*: The amount of time the stimulator is delivering current compared with the rest period between contractions, usually measured in seconds
- *Ramp*: The time in seconds from when the current begins to the peak current (e.g., 3-second ramp up, 6-second contraction, 2-second ramp down)
- *Polarity*: Electrode may be either the anode (+) or cathode (–) (not relevant when using AC)

Source: Adapted from Levine and Bockstahler (2014).

Figure 16.1 A portable NMES unit on a patient. The accuracy of electrode placement is affected by the mobility of the skin at the treatment site – even an electrode taped to skin may move with the skin and not stay over the desired muscle.

inhibit muscle activity. For example, during gait training or standing exercises, the required movement assisted by muscle recruitment from electrical stimulation may be rapid enough to be completed before the peak amplitude is reached (Knaflitx *et al.*, 1990). A patient does not get adequate benefit from increased force of contraction when it is needed. In contrast, a patient with spasticity may need a longer ramp time of 4–8 seconds to avoid stimulating an excess increase in tone (Cameron, 2003). Ramp is not a parameter applicable to continuous flow forms of electrical stimulation. Many clinicians start with duty cycle ratios between 1:2 and 1:5 and watch for signs of fatigue, which indicates the need for a longer off-time (Levine and Bockstahler, 2014).

Patient Preparation

It is advisable to clip the patient's hair for optimal contact of electrodes with skin, although in short-haired patients this may not be necessary. The clip patches also act as a guide for electrode placement in the case of home therapy being assigned using portable electrical stimulation units. The accuracy of electrode placement on clipped patches does also depend on the mobility of the skin at the treatment site – even an electrode taped to skin may move with the skin and not stay over the desired muscle. A coupling medium is necessary to transmit the electrical current from the electrodes to the tissues. Some electrodes are carbon silicon-rubber and need to be used with an aqueous gel, while others are coated with a conductive polymer. Commonly used coupling media include gels, moistened sponges, or paper towels; sponges and paper towels tend to dry out, and rewetting is necessary every 30 minutes. Conductive performance of any electrode decreases over time. Electrodes should be of the appropriate size to stimulate the desired muscle without stimulating unwanted muscles. The smaller the electrode, the higher the current density that enters the muscle, and the more uncomfortable the stimulus may be (Figure 16.1).

Precautions and Contraindications of Electrical Stimulation

The safety of the clinician and patient is of utmost importance when working with animals. A patient who is surprised by a sudden muscle contraction or who is painful may respond with a bite, scratch, or a kick. Most canine patients have bite inhibition and will provide warning, however it is wise to provide adequate restraint and to keep away

Figure 16.2 The use of NMES following hemilaminectomy surgery. Notice that a rolled towel may be helpful to support the patient if they can stand.

Figure 16.3 Looking down on the placement of the conduction pads. Notice the tube of gel in the right corner of the image that is used for the coupling medium.

from the patient's head (or species-appropriate danger zone). It is our obligation in veterinary medicine to understand and read patient stress, to minimize stress and pain, and to err on the side of caution rather than administering a painful therapy (Niebaum, 2013). It cannot be emphasized enough to utilize low-stress handling techniques such as those developed by Dr. Sophia Yin (Yin, 2009).

Caution must be used always when treating a patient with electrical stimulation because of the potentially harmful effects it may have if applied inappropriately (Figure 16.2). The canine patient may not be able to communicate the intensity of an electrical stimulus or completely sense the stimulus in situations where neurologic impairment is present (Figure 16.3). Burns from electrical stimulation have been reported and are associated with too high levels of stimulation and usually over areas if diminished sensation or high adipose content. Thus, precautions for the use of electrical stimulation need to be considered when treating areas with diminished sensation, directly over open wounds, areas of dermal irritation, and over areas of high fat content (Figure 16.4) (Hecox *et al.*, 1994; Cameron, 2003; Niebaum, 2013).

Figure 16.4 Initial placement of the electrical pads prior to additional tape placement. Notice positive (black wire) and negative pads (red wire) are placed opposite each other.

One contraindication of electrical stimulation is stimulation directly over the carotid sinus in the neck just behind the vertical ramus of the mandible, or over the pharyngeal area. Stimulation to this area may induce a rapid fall in blood pressure, which could cause syncope. Another area where the placement of electrical stimulation is contraindicated is over the heart. Electrical stimulation in patients with pacemakers may interfere with the functioning of the pacemaker device, altering heart rate or rhythm. Likewise, electrical stimulation should not be used in patients with seizure history as the electrical currents may induce a seizure episode. Since the effects of electrical stimulation to the developing fetus and pregnant uterus are unknown, use of electrical stimulation over the trunk, abdomen, low back, hips, or pelvis during pregnancy is contraindicated. Areas of thrombosis, suspected thrombosis, or thrombophlebitis should be avoided because of the potential risk of the embolus releasing (Box 16.3). This is of importance in cats with rear limb pain or weakness and without a definitive diagnosis; femoral arterial blood flow should be verified with diagnostic ultrasound before considering using electrical stimulation in a cat with weak or painful rear limbs. Also, considering the circulation changes associated with electrical stimulation, it is contraindicated to use

Box 16.3 Indications and contraindications for electrical stimulation

Contraindications

- Pregnancy
- Wounds
- Malignancy
- Pacemaker patients
- Over laminectomy site
- Seizure patients
- Areas of thrombosis
- Pharyngeal area
- Carotid sinus
- Where there is movement (i.e., fracture)

Indications

- Increasing muscle strength
- Increasing range of movement (ROM)
- Decreasing acute pain
- Decreasing chronic pain
- Reducing muscle spasm
- Promotion of soft tissue healing
- Promotion of fracture healing

Source: Adapted from Hanks *et al.* (2015).

over areas of infection because of the theoretical potential to cause sepsis. Another contraindication concerning the circulatory effect is over a malignancy, though occasionally electrical stimulation is used to control pain in patients with malignancies when quality of life outweighs the possible risks of the treatment (Hecox *et al.*, 1994; Cameron, 2003; Niebaum, 2013). The use of electrical stimulation should be avoided in any situation where active movement of the area is contraindicated, such as over an unstable fracture. Electrical stimulation should never be used over the eyes.

Adverse Effects

There are very few adverse effects of electrical stimulation, but electrical burns, skin irritation, and pain can occur if the electrical stimulation is applied inappropriately. If there is not enough conduction medium, the risk of a burn increases due to inadequate conduction in to the tissues. Electrical burns are most common with the use of direct current due to the constant delivery of the current. Skin irritation can occur if the patient has skin sensitivities or is allergic to the contact surface of the electrodes, which is rare. In this case a different type of electrode can be tried. Some patients find electrical stimulation painful. In those cases, decreasing the pulse width, increasing the ramp time, and/or increasing the intensity slowly over time may help their tolerance. If it is still painful, other forms of treatment should be utilized (Cameron, 2003).

Neuromuscular Electrical Stimulation for Muscle Strengthening

Neuromuscular electrical stimulation (NMES) is a non-pharmacological treatment method that can be used to address muscle weakness or atrophy with patients who are unwilling or unable to actively contract a muscle. It is usually applied at higher frequencies (20–50 Hz) with the goal being to achieve muscle tetany and contraction that can be used for functional purposes (Valenti, 1964; Tacker *et al.*, 1991). The goal is to provide an electric potential to a muscle to elicit a non-voluntary contraction. The electrical current creates a muscle contraction by depolarizing the motor nerve (Niebaum, 2013).

Adverse changes in muscle strength and function have been reported following injury or surgery. The primary injury alone will create changes in muscle activity and motor control that are initially thought to be protective. However, unless specifically addressed, these changes in motor control will persist long after pain has resolved and tissue healing has taken place (Ichihara *et al.*, 2009). Surgery and its resultant inflammation during healing time can further compromise muscle activity and motor control (Ocal and Sarierler, 2007). These changes include reduced muscle contractility, delayed onset of contraction (poor muscle timing), and atrophy of muscle tissue (Ocal and Sarierler, 2007; Niebaum, 2013). This inhibition of muscle function, if left unattended, has been shown to contribute to injury of an area (whether repeat injury or injury to other body parts). If muscle activity is decreased because of neurological compromise from injury or surgery, the muscle will atrophy and significant loss of strength and fine motor control will occur. Changes at the cellular level also occur in muscles that atrophy; atrophy is particularly rapid when neurogenic. A reduction in mitochondria, glycogen stores, enzymes associated with muscle building, and the number of functioning muscle cells is reduced, leaving a smaller, metabolically deficient muscle unit. The patient will be at a disadvantage when trying to return to normal function, even if resolution of the neurological compromise occurs. Many of the functional problems associated with a neurologically impaired patient can be found to be due to the detrimental changes associated with a lack of normal exercise. If the muscle is not used as much, some atrophy ensues.

Box 16.4 Current parameters for strengthening

- *Frequency*: Generally between 25 and 50 Hz (these have been shown in humans to produce strong tetanic contractions while minimizing fatigue). An alternative in dogs that are not very tolerant of electrical stimulation is to use a low-frequency (<10 Hz). This will provide a twitch contraction and is better tolerated in some cases
- *Waveforms*: Many waveforms exist; any waveform capable of depolarizing the muscle is acceptable
- *Pulse or phase duration*: Between 150 and 250 μs
- *Ramp up/down (rise and decay time)*: Adjust between 2 and 4 s up to increase comfort; 1–2 s down
- *On/off time*: A 1: 3 or 1: 4 ratio; 10 s on, 30 or 40 s off is commonly used. This may be decreased as muscle strength improves. A 1: 1 or 1: 2 ratio is usually used for muscle endurance training. To prevent atrophy 1:1 (15 s on:15 s off) (Gersh, 1992)
- *Treatment time*: 10–20 minutes
- *Frequency of treatment*: 3–7 sessions/week
- *Amplitude*: Sufficient to cause a strong muscle contraction
- *Position*: Initially, the dog may be more comfortable in a relaxed position but as it becomes more familiar with the sensation, using it in a functional position may result in improved strength gains (Niebaum, 2013)

Source: Adapted from Levine and Bockstahler (2014).

These changes in the muscle can be found even in animals without neurological compromise (Box 16.4). Immobilization, such as splinting during the healing time of a fracture, causes atrophy and its associated consequences.

Since NMES can stimulate muscle contraction without patient contribution (Gruner, 1986), if the patient is unable to maximally contract a muscle, an electrically induced muscle contraction may create a bigger "top off" force and therefore improve muscle activity over patient activity alone (Figure 16.5) (Fitzgerald *et al.*, 2003). However, studies have shown that electrical stimulation is not superior to voluntary strength training, and is more effective when coordinated with voluntary muscle activity (Currier and Mann, 1983; Laughmann *et al.*, 1983; McMiken *et al.*, 1983; Selkowitz, 1985; Wolf *et al.*, 1986). A systematic review on the effects of NMES applied to the quadriceps muscle following anterior cruciate ligament reconstruction in human patients showed that combined with

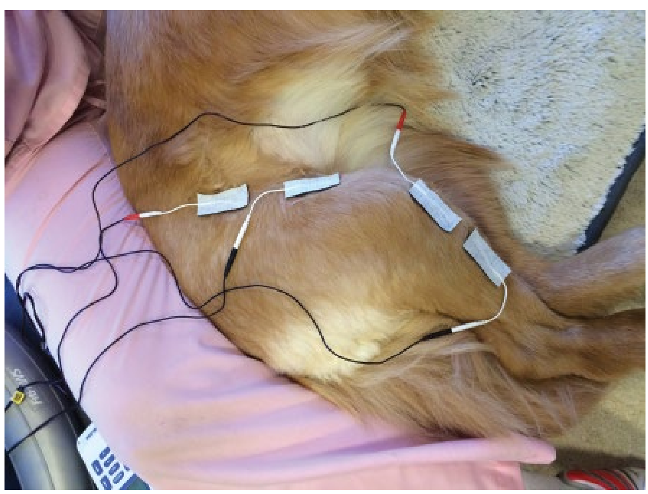

Figure 16.5 Shows use of pads that do not require taping because they have a sticky adhesive on their back.

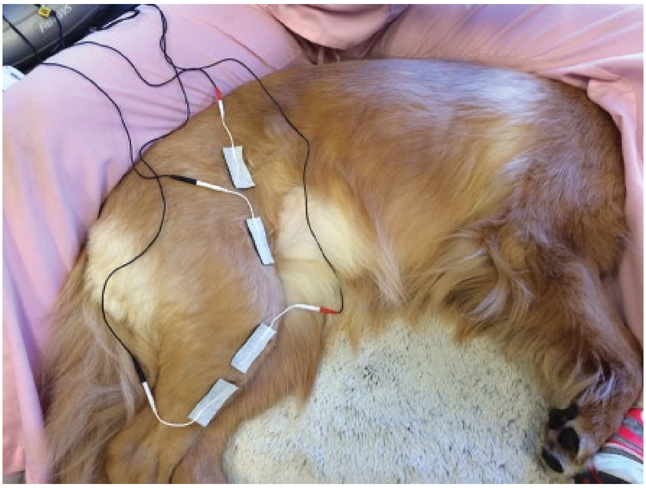

Figure 16.6 Additional view of pad placement without taping.

exercise, NMES may be more effective in improving quadriceps strength than strengthening through exercise alone (Figure 16.6) (Kim *et al*., 2010). A study of dogs with experimentally induced cranial cruciate tears was performed with the dogs undergoing NMES 3 weeks after surgical stabilization. Those dogs had significantly better lameness score and larger thigh circumference than control dogs (Johnson *et al*., 1997).

Neuromuscular Electrical Stimulation for Edema

NMES can also help to decrease tissue edema. Muscle contractions during normal movements provide a pumping action, which helps with venous return (Naamani *et al*., 1995). Weakness, pain, trauma, bed rest, or other neuromuscular issues can interfere with venous return. Muscle pumping can be achieved either by using low-frequency (10 pps) twitch-like contractions on a continuous setting or tetanic contractions (frequency greater than 25–30 pps) with a rapid on–off cycle (less than 5 seconds on-time). Electrical stimulation applied to the affected muscles may also generally improve circulation to a compromised area by increasing muscle pumping action (Figure 16.7). However, the induced muscle contraction should only be strong enough to produce an

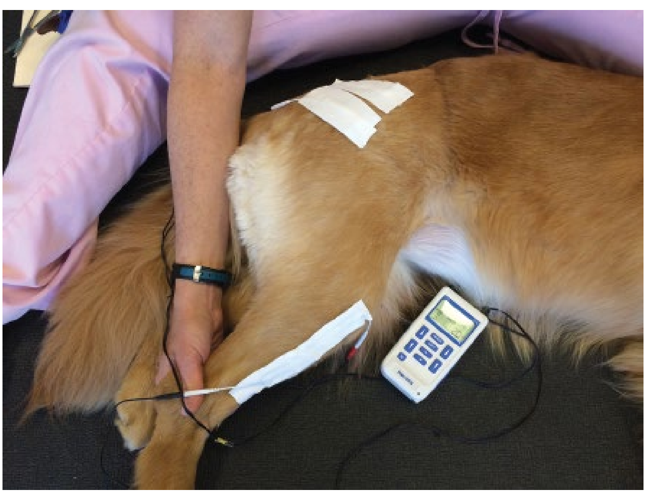

Figure 16.7 Pad placement along stifle and hip.

Box 16.5 Electrical stimulation parameters for edema

- *Frequency*: Between 35–50 Hz or pps
- *Pulse duration*: Between 150 and 200 ms for small muscles and 200–350 ms for large muscles
- *Ramp up and down time*: At least 1 s
- *On/off time*: Equal on and off times or 1:1 ratio
- *Treatment time*: 30 minutes
- *Frequency of treatment*: 2 times/day
- *Amplitude*: To visible contraction (Griffin *et al.*, 1990)
- *Position*: How the dog is most comfortable and preferably with the affected area elevated

initial, low-level, visible contraction since strong contractions may impede venous blood and lymph flow (Lake, 1992).

When using electrical stimulation to reduce edema it is best applied to major muscle groups around the edema to promote reciprocal contraction and relaxation of the agonist and antagonist muscles (Box 16.5). This alternating contraction will aid in the pumping mechanism that venous return relies upon. The electrode pads can also be placed proximal and distal to the affected area with the same type of intermittent stimulation (Hecox *et al.*, 1994). Studies have shown significant edema reduction after a single 15- to 30-minute treatment in human patients (Lake, 1988; Griffin *et al.*, 1990).

Electrical Stimulation for Wound Healing

The use of electrical stimulation is indicated to assist in wound healing. It is best to ask the supervising veterinarian before applying this technique over any wounds. Significant caution needs to be taken when applying electrical stimulation over areas of diminished sensation. The person carrying out the therapy should have a good understanding of the intensity of stimulation provided over a region with neurological compromise. The intensity of electrical stimulation units can be turned up far beyond what may be tolerable to a neurologically intact veterinary patient. High-intensity electrical stimulation will cause burns to the skin. Different waveforms will be tolerated at different intensities. The veterinary technician should have a good understanding of what a specific waveform and intensity feels like before placing it on a patient with sensory compromise to ensure an appropriate and safe intensity is used for treatment.

Box 16.6 Parameters for wound healing

- *Frequency*: 60–125 Hz or pps
- *Pulse duration*: 40–100 ms
- *Amplitude*: High enough to elicit a comfortable sensory response
- *Treatment time*: 45–60 minutes
- *Polarity*: Negative if treating infected area, positive if treating clean area (Cameron, 2003)

Remember that adipose tissue is a poor conductor of electricity. This lack of conduction through the fat cells will cause heat to build up in the tissues, predisposing these areas to a higher likelihood of an adverse event occurring with the use of electrical stimulation (Hecox *et al.*, 1994; Cameron, 2003; Niebaum, 2013).

Electrical stimulation at the site of a wound has been shown to be effective in promoting tissue healing by increasing local circulation, retarding bacterial growth, and stimulating tissue repair mechanisms (Box 16.6). It is believed to aid the wound-healing process by mimicking natural electrical currents in the skin that occur when skin is injured (Foulds and Barker, 1983). The current stimulates and directs migration of macrophages and neutrophils (Eberhardt *et al.*, 1986) and stimulates fibroblasts (Alvarez *et al.*, 1983). Many different types of electrical current have been shown to improve wound healing. In 1999, a meta-analysis was done on the effects of various waveforms of electrical stimulation on chronic wound healing in human patients. The clinical trials analyzed the use of four types of currents: low-intensity direct current (LIDC), alternating current (AC), high-voltage pulsed current (HVPC), and transcutaneous electrical nerve stimulation (TENS). The results showed that all types of electrical stimulation utilized were found to be an effective adjunctive therapy for chronic wound healing (Gardner *et al.*, 1999).

Physiologically, the effects of electrical stimulation on wound healing can be explained by the electrical properties of the skin. After an injury to the skin, an LIDC has been found to occur, which is thought to contribute to promoting the healing response found in healthy, injured skin tissue (Nuccitelli, 2003; Ojingwa and Isseroff, 2003). Electrical stimulation is thought to be able to magnify this potentiated normal endogenous biological healing response in the skin, promoting tissue proliferation and improving the rate of healing (Nuccitelli, 2003; Huttenlocher and Horwitz, 2007). Improvements in wound healing with the use of electrical stimulation have been demonstrated in full thickness wounds (Carely and Lepley, 1962) and surgical incisions (Taxskan *et al.*, 1997; Demir *et al.*, 2004) in animals as well as when used on pressure ulcers (Reger *et al.* 1999), burns, diabetic ulcers (Greenberg *et al.*, 2000; Thawer and Houghton, 2001), and ulcers associated with ischemia (Morris *et al.*, 2009).

Electrode Size and Placement for Neuromuscular Electrical Stimulation

Correct electrode placement is crucial to obtain the best muscle contraction. The technique is to place electrodes over the belly of the muscle(s) to be stimulated, with one of the electrodes placed over a motor point. The motor point is the point over the muscle belly at which the smallest amount of current is required to produce the muscle contraction (Lake, 1992). The distance between electrodes influences the depth and course of the electrical current. The greater the distance between electrodes, the deeper the current travels, and conversely the closer together the electrodes are the more superficially the current travels and so less muscle fibers are stimulated (Cameron, 2003). It may be beneficial to have different types of electrodes in the clinic in case a patient demonstrates an allergy to a certain electrode or there is difficulty keeping one type adhered to the patient. In dogs with thicker coats, the fur needs to be clipped or shaved for adequate contact (Niebaum, 2013). The authors prefer to use tape or flexible bandaging to hold the electrodes in place to maintain good contact with the skin throughout the treatment, especially if the patient moves (Box 16.7).

The size of the electrodes should match the muscle(s) being targeted. Small electrodes are available to accommodate smaller treatment areas, but the current density produced over a small surface area can cause discomfort when used at higher treatment amplitudes. In contrast, electrodes that are too large can cause the current to spread to other muscles. In addition, the electrodes, and their coupling medium, should not encounter each other, as this will result in the current flowing directly from one electrode to the other instead of in to the targeted muscle tissues (Lake, 1992; Niebaum, 2013).

Box 16.7 Criteria for electrodes

The main criteria in choosing electrodes are as follows:

- Flexible enough to conform to the tissue
- May be trimmed to a specific size
- Have a low resistance
- Are highly conductive
- May be used many times
- Are inexpensive

Source: Adapted from Levine and Bockstahler (2014).

Electrical Stimulation for Pain Control

When used for pain control, TENS can be a beneficial modality. It is an alternate form of electrical stimulation that has been historically administered at high frequencies to provide pain relief (Deyo *et al.*, 1990) but is now used at very low frequencies (2–10 Hz) (Sluka and Walsh, 2003). Low frequency stimulates sensory nerves instead of motor nerves, which is thought to reduce pain perception through the gate control theory of pain inhibition (Melzack and Wall, 1965; Niebaum, 2013). The gate control theory is based on the idea that stimulation of large myelinated nerve fibers, which transmit signals rapidly into the dorsal horn of the spinal cord, can interfere with the slower transmission of pain nerve (nociceptors) to higher centers where pain is perceived. According to the theory, the impulses from TENS stimulate fast conducting mechanoreceptors (Cameron, 2003). Pain signals are transmitted by small cutaneous A-δ and C fibers. If a TENS current is applied, the large cutaneous (A-β) fibers are stimulated (Levine and Bockstahler, 2014). The signals from the A-β fibers activate inhibitory interneurons in the substantia gelatinosa of the spinal cord dorsal horn and block the transmission of pain impulses from the periphery to the brain (Melzack and Wall, 1965). Additional mechanisms involve the release of endogenous opiates (Levine and Bockstahler, 2014). Pain relief is provided only while the stimulation is being delivered. However, it may be beneficial for patients when pain is interfering with their abilities to perform therapeutic activities and functional movements (Niebaum, 2013) (see Chapter 3). Any deactivation of nociceptors can help to reduce the general state of wind up and so over time TENS can aid in reducing CNS sensitization. The presence of pain alone has been shown to cause changes in muscle activity and to subsequently decrease muscular forces throughout the whole body. A reduction in pain will result in an improvement in function. TENS is an excellent adjunctive pain therapy treatment that can be used in combination with pharmaceuticals in acute pain situations, such as following trauma or surgery, as well as in chronic pain situations associated with degenerative change (Box 16.8).

Box 16.8 Recommended parameters for TENS

- Amplitude high enough to elicit a comfortable sensory response
- Pulse duration between 50 and 100 ms
- Frequency between 30 and 150 Hz
- Duration of treatment varies according to the activity
- Placement of electrodes: over or around the area of pain, over the peripheral nerve or spinal nerve roots that innervate the painful region, or over acupuncture points

Source: Adapted from Nolan (2005).

Pulsed Electromagnetic Field Therapy

Pulsed electromagnetic field (PEMF) therapy signals are the most commonly used electromagnetic signals for therapeutic applications. This modality works through the creation of pulsed magnetic fields that have been found to have therapeutic benefit in the healing responses of various tissues. PEMF therapies provide therapeutic microcurrents using targeting signals to enhance the natural regenerative pathways used by the body (Seegers *et al.*, 2001; Pilla, 2003). These signals come in the form of an electromagnetic field. A magnetic field is the region surrounding a magnet or an electric current. Electric current delivered through a wire to a magnet creates the pulsed magnetic field, which in turn induces an electric current that affects the targeted tissue to create a beneficial healing response (Pilla, 2003). This is termed inductive coupling. The ability to influence tissues remotely can enhance healing of the target tissue while penetrating through more superficial tissues (for example stimulating bone healing in closed fractures) and even penetrating bandages to stimulate repair in wound beds (Trock, 2000).

There is increasing use of PEMF as adjunctive therapy for a multitude of musculoskeletal injuries and problems. In current clinical veterinary practice, electromagnetic field (EMF) modalities have been shown to have therapeutic benefits for bone and wound repair and acute and chronic pain relief (Pilla, 2006). It is a safe, simple, and cost-effective modality with minimal to no side-effects, and is therefore becoming a popular modality choice in veterinary rehabilitation.

Pulsed Electromagnetic Field Therapy for Wound Healing

The treatment of wounds can be costly, require extensive care, and be time consuming. It may be challenging for owners to consistently bring their dogs to a veterinary practice regularly for wound care due to these factors. A modality applied at home that could assist in the healing wounds, especially in high-risk populations for delayed healing, would be advantageous to the pet owner, the veterinary staff, and the patient. Radiofrequency PEMF has been shown to enhance healing of skin wounds (Pilla, 2006). In a study done in 2007, the authors successfully demonstrated that treating wounds with PEMF with very specific settings accelerated early diabetic wound healing in rats. There was increased wound tensile strength at 21 days. The wounds were exposed to a 1.0 G PEMF signal for 30 minutes twice daily for 21 days (Strauch *et al.*, 2007). EMF strength is expressed in gauss units (symbol G). Wound healing in diabetic animal populations is enhanced in the early stages by PEMF. This is accomplished through an increase in myofibroblastic proliferation resulting in significantly enhanced wound closure and re-epithelialization 10 days after wound onset (Cheing *et al.*, 2014). The advantage of using PEMF over direct contact electrical stimulation to accelerate wound healing can be found in the production of an EMF created by this modality. This field allows the application of electric current to be somewhat remote from the wound bed, allowing the electromagnetic signals to penetrate bandaging and dressing material to affect the underlying tissues of the wound.

Pulsed Electromagnetic Field Therapy for Bone Healing

Understanding how PEMF contributes to bone healing requires some knowledge of the normal processes involved in bone growth. In response to stress, bone deforms slightly. Being a crystalline structure, compression in a long bone will be subject to the piezoelectric effect. The piezoelectric effect states that compressing a crystalline structure results in negatively charged electrons migrating to the side of concavity. It is thought that the electronegative potential creates an environment where osteoblasts can function normally but osteoclast activity is halted. Thus, the formation of bone tissue exceeds the normal physiologic breakdown of tissue for repair, which results in net bone growth. This is the reason why the trabeculae of the bone are organized in a manner consistent with the stress placed on the bone. This results in trabecular patterns that follow strain patterns in bone and causes greater bone mass to be present in areas of greater physical stress.

PEMF has been shown to increase healing time in non-union and delayed-union fractures. In these cases, it has been theorized that there is an absence of electrical charge to help stimulate osteoblastic activity. The application of an electrical charge applied to the bone using PEMF has been shown to assist in bone healing. A study in 2012 showed that PEMF stimulation is an effective non-invasive method for improving non-infected tibial union abnormalities. The time of treatment onset did not affect the incidence of fracture healing. Longer treatment duration showed a trend of increased probability of union (Assiotis *et al.*, 2012). In general, relevant clinical responses to PEMF are not immediate in delayed-healing bone, requiring daily treatment for sometimes several months in the case of non-union fractures (Pilla, 1993). A recent systematic review and meta-analysis found that PEMF may not be beneficial in decreasing the incidence of non-union fractures, however the authors did find a decreased healing time in acute fractures and diaphyseal fractures when PEMF was used (Hannemann *et al.*, 2014). PEMF is approved by the US Food and Drug Administration (FDA) for use as an adjunctive treatment to aid in fracture repair (Canapp, 2007; Galkowski, *et al.*, 2009).

Pulsed Electromagnetic Field Therapy for Osteoarthritis

Osteoarthritis has been estimated to affect 20% of the US canine population in middle age and 90% of older dogs have osteoarthritis in one or more joints; the percentages are even higher for the human population (Budsberg, 2010). This widely referenced estimate, in practical terms, translates to over 10 million dogs. In cats, a report indicated that a 26% radiographic prevalence of appendicular osteoarthritis and a 90% prevalence of all types of degenerative joint disease, but the study only included cats older than 12 years of age (Hardie *et al.*, 2002; Bennett *et al.*, 2012). Thus, the identification and management of the disease is of the utmost importance to the small animal clinician. Clinical signs may include stiffness, fatigue, lameness, pain, difficulty transferring from one position to another, and the inability to perform daily activities such as jumping on to the couch or climbing stairs. It can affect a dog's quality of life, and in severe cases may cause aggression, anorexia, and inability to ambulate.

Non-invasive therapies aimed at ameliorating symptoms of osteoarthritis typically include weight management/loss, physical rehabilitation, and the use of non-steroidal anti-inflammatory drugs (NSAIDs) along with other pharmaceuticals. Recent research into modalities and nutritional supplements aimed at modifying the disease are gaining acceptance as part of a multimodal approach. One such treatment for osteoarthritis is the application of PEMF therapy to the joint and periarticular tissues (O'Sullivan *et al.*, 2013).

In 2013, a randomized, controlled clinical trial was published that evaluated the efficacy of pulsed signal therapy (PST), which is the application of PEMF, in dogs with osteoarthritis, as measured by the Canine Brief Pain Inventory (CBPI) Severity and Interference scores (O'Sullivan *et al.*, 2013). The CBPI is a validated survey instrument to assess chronic pain in dogs (Brown *et al.*, 2008). The results showed the PST group performed significantly better than the control group, but the objective measures of range of motion and peak vertical force were not statistically significant between groups receiving and not receiving PEMF. It was speculated that the range of motion may have been limited more by mechanical means (i.e., osteophytes, periarticular fibrosis) than by pain (O'Sullivan *et al.*, 2013).

In 2009, a meta-analysis of randomized controlled trials was done to determine the effectiveness of PEMF in the management of osteoarthritis of the knee in humans. It concluded that PEMF improved function and clinical scores in patients with osteoarthritis of the knee and should be considered in conjunction with other treatments in the management of their disease process (Vavken *et al.*, 2009).

The theory behind the mechanisms of the effects of PEMF on osteoarthritic pain lies in the anabolic effects that PEMF has on osteoblast and chondrocyte proliferation, which produces healing effects at the cellular level (Diniz *et al.*, 2002) by enhancing regeneration of articular cartilage (Ciombor *et al.*, 2003). Pain in osteoarthritis is perceived by the patient from the stimulation of unmyelinated and small myelinated nerve fibers in the joint and the surrounding tissues of the joint (joint capsule, ligaments, synovium) (Felson, 2005). In addition, central sensitization, which is a hyperexcitability of neurons in the central nervous system associated with chronic pain conditions, results in changes within the brain responsible for pain perception, patient affect, cognition, and sensorimotor function (Felson, 2005; Imamura *et al.*, 2008). The literature provides mixed clinical evidence in the support of PEMF in pain reduction in human patients who suffer from osteoarthritis pain. A systematic review with meta-analysis was published in 2013 and showed PEMF improved physical function in people with osteoarthritis of the knee but did not result in significant improvement in pain levels (Negm *et al.*, 2013). Osteoarthritic pain originates from the joint capsule and subchondral bone, not cartilage.

There is currently a lack of quality studies in the literature investigating the effects of PEMF on osteoarthritis-related pain. It is difficult to adequately determine how well PEMF effectively works on osteoarthritic pain, although some evidence exists that supports an improvement in the function of patients with osteoarthritis if PEMF is used (Hunter, 2011).

Treatment Protocol for the Assisi Loop®

The Assisi Loop (www.assisianimalhealth.com/) offers targeted PEMF in a portable, easy-to-use device that can be used in clinic or in the home/stable (Figure 16.8). The Loop has been proven to reduce pain and inflammation and has been cleared by the FDA for palliative treatment of postoperative edema and pain in humans (Rohde *et al.*, 2015).

Therapy starts with a minimum of 3–4 15-minute treatments per day for acute and chronic or degenerative conditions. When treating acute conditions with an Assisi Loop-Automatic, it is recommended that treatments begin immediately and that the Loop is left on automatic mode for anywhere from 48 hours to two weeks. This depends on the animal's progress. When treating following surgery, treatment can begin immediately after the operation and should continue until the surgical site is healed.

Figure 16.8 Image of an Assisi Loop used for targeted PEMF.

For chronic or degenerative conditions, the company recommends that you continue with 3–4 treatments per day for 7–10 days and monitor the animal until you see improved mobility and less pain response. Taper down to 1–2 treatments per day or even 1–3 treatments per week. With some chronic and degenerative conditions, the patient may get to the point that they would only be treated as needed for pain, particularly if it is a condition that is prone to flare-ups.

There are no contraindications with other modalities, and the Loop is a take-home treatment for pet owners to continue healing between rehabilitation visits. However, given that the Loop and other treatment modalities, such as laser, have different mechanisms of action, the company recommends that you allow 2 hours between treatments to be certain that each treatment can operate at its fullest efficacy.

Using the Loop over a metal implant or brace will not cause any harm, but it may distract or weaken the signal. If possible, position the Loop so that it is at a 45° angle from the metallic implant. If you are not able to do this, then you may consider using the Loop more frequently. The FDA recommends against using the human version of the Loop over tissue known to contain implanted metallic leads such as pacemakers.

Targeted PEMF has no contraindication for the treatment of patients with cancer or for post-cancer surgical healing. Much of the human clinical trial work that has been done has been for patients following mastectomy reconstruction.

Pulsed Electromagnetic Field Therapy Bed or Blanket

PEMF uses pulsing magnetic fields developed by pulsing a small amount of battery current through coils of wire to initiate normal biological cellular reactions that result in improved circulation and provide pain relief. Respond Systems™ makes three products to provide pain relief for dogs and cats from common problems associated with hip dysplasia, arthritis, muscle, tendon or ligament injury, and old-age stiffness and soreness. For horses, they make this product as a blanket, wraps, and mat for the horse to stand upon (https://respondsystems.com/pemf/how-it-works/).

Conclusions

Selection of the appropriate modality depends largely on an understanding of the diagnosis, an accurate assessment of the stage of tissue healing and repair, an accurate clinical assessment of the functional limitations, the established treatment goals, and continued reevaluation of the patient (Pinna *et al*., 2013). electrical stimulation is most commonly used to reduce pain and muscle spasm, in both acute and chronic circumstances, and to increase muscle strength. PEMF is used for healing of bone and tissues and pain management.

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17

Modalities Part 4: Therapeutic Ultrasound

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Introduction

Therapeutic ultrasound uses sound waves generated by piezoelectric effect at frequencies greater than 20 000 Hz to stimulate tissues and create physiologic effects. These effects include stimulation of fibroblast activity, improvement of blood flow, increase of protein synthesis, increase of tissue extensibility, decreased pain, and stimulation of soft tissue and bone healing (Steiss and McCauley, 2004; Baxter and McDonough, 2007; Niebaum, 2013). Therapeutic ultrasound devices may use short bursts of sound waves or a continuous stream to deliver ultrasonic energy to tissues. This modality has been used since the 1950s for indications such as tendon injuries and bursitis (Miller *et al.*, 2012). Based on a National Library of Medicine search, the first clinical report of its use in veterinary medicine was in 2009 (Mueller *et al.*, 2009).

Physical Principles

Sound Waves

Sound waves used for therapy consist of mechanical vibrations that are propagated longitudinally into the biologic tissues via a means of a transducer that is placed directly in contact with patient's skin. As sound moves through tissues, energy is absorbed and

Physical Rehabilitation for Veterinary Technicians and Nurses, First Edition.
Edited by Mary Ellen Goldberg and Julia E. Tomlinson.

converted to kinetic energy (Niebaum, 2013). Sound waves create pressure and tensile forces on biologic tissues causing them to oscillate. The frequency of the sound waves determines the degree of sound wave oscillation.

Absorption of Ultrasound Waves

Factors that affect the travel and absorption of sound waves includes the substance traveled through to the target tissue, the components of the target tissue, the coupling media, and the treatment parameters. In general, sound is likely to move faster through solids, slowest through gases. Therefore, ultrasound transmission is impeded by air between transducer and skin. In order to reduce sound reflection at the skin resulting from air pockets, it is necessary to use a coupling medium. Liquids such as water and alcohol are suitable but evaporate quickly. A coupling gel is better for direct contact of probe on skin. Gels can be water or lipid based, but the latter are hard to remove after treatment, so water-based gels are more commonly used. Indirect coupling is when the probe does not contact the skin but moves through a medium, most commonly water. This can be useful for applying therapeutic ultrasound over irregularly shaped areas when complete contact with even the smallest treatment probe is not possible.

The patient's body consists of a variety of tissues with different acoustic impedances. Ultrasound is even reflected or absorbed differently by each structure within a tissue. Absorption of energy is greatest in tissues with high protein content (e.g., bone) and relatively low in adipose tissue (Steiss and McCauley, 2004). The prescribing veterinarian and the therapist must be familiar with the composition of the target tissue, and the tissues that the ultrasound waves have to travel through in order to reach the target tissue. Because hair has a high protein content it can diminish ultrasound penetration and therefore hair coat should be clipped very short, or even shaved prior to ultrasound therapy (Steiss and Adams, 1999).

Mechanical Effects of Ultrasound Waves

Cavitation is the most commonly cited mechanical effect of therapeutic ultrasound. It is the formation of small gas bubbles in tissues that occur when ultrasonic waves vibrate (Bockstahler *et al.*, 2004). Ultrasonic cavitation depends on the pressure amplitude of ultrasound waves; ultrasound transmitted into a tissue may cause pressure amplitudes of several megapascals. The resultant tensile stress is supported by the tissue and negative tension in the tissue can be several times atmospheric pressure; even diagnostic ultrasound can affect tissues this way (Miller *et al.*, 2012). Cavitation can affect tissue metabolism by causing temperature elevation, mechanical stress, or free radical production (O'Brien, 2007).

The different forms of cavitation are as follows:

- *Stable cavitation* is when the gas bubbles remain intact and move with the ultrasonic flux. This form of cavitation is desired to produce therapeutic effects.
- *Transient cavitation* is when the gas bubbles rapidly expand and collapse resulting in high pressure and increased temperature which can damage tissues. It primarily causes local tissue injury in the immediate vicinity of the cavitational activity, including cell death and hemorrhage. To prevent transient cavitation, avoid using a high-intensity setting especially with continuous wave ultrasound.

Cavitation is a secondary effect of transmission of ultrasound waves in tissues, another secondary effect is acoustic streaming, which is flow of currents in tissue fluid. The direct mechanical effects of ultrasound waves are compressional, tensile, and shear stresses; when an ultrasound wave moves through tissue, a mechanical stress is induced. We know from Wolff's law that bones remodel in response to stress, and we know that tendon cells produce collagen in response to high-frequency vibrations

(Thompson *et al.*, 2015) but we do not know how sustained a mechanical stimulus must be in order to achieve a therapeutic response from ultrasound. The therapist aims to use the mechanical effects of ultrasound to stimulate breakdown of abnormal tissue (including scar tissue and mineralization) and to stimulate a tissue repair response. The therapist must also avoid adverse effects (too much tissue breakdown and disruption). Clinical research determining ideal settings (depth, intensity, time) for treatment of different pathologies is still lacking in both human and veterinary medicine. We do know that the mechanical effects of ultrasound act on the cell membrane, causing mild disruption and increased permeability. Ultrasound has been shown to result in a number of intracellular events that include triggering biochemical reactions and changes in gene expression. This can result in increased cell proliferation or turnover (Furusawa *et al.*, 2014). Some examples of tissue responses are vasoconstriction, ischemia, extravasation, reperfusion injury, and immune reaction (Miller *et al.*, 2012).

Thermal Effects of Ultrasound

Absorption of ultrasound waves by a tissue represents a portion of the wave energy that is converted into heat. If heat is produced by this process at a faster rate than can be removed/dissipated then a temperature increase will occur in the tissue. The use of unfocused heating can be moderated to produce enhanced healing without injury (Miller *et al.*, 2012). Ideally, tissues should be heated by 4–5 °C for improving tissue flexibility (Levine *et al.*, 2001). As therapists, we may want to increase local tissue temperature at our target in order to increased enzyme-based reactions, though there is a ceiling effect beyond which too much heat can denature enzymes (O'Brien, 2007). Heat can partially denature (unfold) collagen by breaking up hydrogen bonds in collagen crosslinks (Chen *et al.*, 1998). This may be a mechanism by which ultrasound stimulates tissue remodeling and breakdown of scar tissue. Alternatively, the heat produced by therapeutic ultrasound can be deliberately concentrated by a focused beam until tissue is coagulated for the purpose of tissue ablation.

Chemical Effects of Ultrasound

The microbubbles of cavitation are under high pressure and temperature just before they collapse. These changes are very fast and so have minimal heating effect on surrounding tissues. However the high temperatures can lead to water breakdown and free radical formation (Furusawa *et al.*, 2014). This "sonolysis" in the presence of oxygen leads to the formation of superoxide radicals, which can stimulate tissue remodeling. Ultrasound results in an immediate elevation of intracellular calcium ions that influx from the extracellular environment.

Treatment Parameters

Frequency

Frequency determines the depth of ultrasonic wave penetration and is measured in hertz (Hz). The frequency selected is based on the tissue to be treated and its depth from the skin surface.

The most common frequencies used in veterinary rehabilitation are 1 MHz and 3.3 MHz. Ultrasound of 1 MHz can effectively penetrate a depth of 2–5 cm, whereas 3.3 MHz can effectively treat tissues at a depth of 1–3 cm (Steiss and Adams, 1999; Levine *et al.*, 2001; Bockstahler *et al.*, 2004).

Intensity

Intensity is the rate of energy delivery per unit area and is measured in watts per centimeter squared (W/cm^2). Intensity is important because it describes the force at which the vibrations are applied (how fast the energy is delivered). Too much force could damage the tissue or, at the very least, result in pain. The higher the intensity, the higher the increase in tissue temperature. When treating an area with thick soft tissue (e.g., the deltoid muscle), a higher frequency should be selected to heat the tissue. In contrast, when treating an area with minimal soft tissue (e.g., the carpal joint), lower intensity and higher frequency are the parameters of choice as denser tissues heat faster. The most common intensities used in veterinary medicine are between 0.1 and 2 W/cm^2.

Duty Cycle

Duty cycle is the fraction of time that sound is emitted during one pulse period. It refers to whether the vibrations are applied continuously over the course of a treatment or whether they are pulsed on and off. The ratio of pulse on to pulse off during a treatment is referred to as the duty cycle. Most devices offer duty cycles in the range of 10%, 25%, 50%, 75%, and 100%. For example, a 10% duty cycle pulsed waveform would have the ultrasound on for a total of 10% of the treatment time and off for 90% of the treatment time. A 100% duty cycle is considered continuous waveform, in which the ultrasound is on 100% of the treatment time. Continuous waveform is used when thermal and non-thermal effects are desired (e.g., muscle spasms and chronic tendinopathies). Pulsed waveform is used when only non-thermal effects are desired (e.g., around bony prominences and in tissues with acute inflammation).

Transducer Heads

There are a variety of sizes of transducer heads available (Figure 17.1) and the size is selected based on the size of the treatment area. The treatment area should be no more than 2–3 times the size of the transducer head (Steiss and McCauley, 2004). Treating an area larger than these parameters decreases the dosage and thermal effects, therefore diminishing the therapeutic effects.

Figure 17.1 A 3 cm sized ultrasound transducer head.

An alternative is to divide a larger area into two smaller treatment areas. Common sizes of transducer heads include 1 cm^2, 3 cm^2, 5 cm^2, and 10 cm^2. The transducer head should be held directly over the treatment tissue and perpendicular to the skin. If it is held at an angle above 75 degrees, the ultrasound will travel along the skin instead of treatment tissue (Michlovitz, and Sparrow, 2012). The transducer head should be moved continuously over the skin at a speed of 4–8 cm/s to achieve uniform distribution of the ultrasound and prevent thermal burning and tissue damage (Michlovitz and Sparrow, 2012).

Coupling Techniques

Water-soluble coupling gel must be placed between the skin and transducer head; otherwise the ultrasound beam is reflected at air–tissue interfaces. Elimination of air as much as possible maximizes tissue penetration of the ultrasound energy (Figure 17.2).

Direct coupling is placement of the transducer head in direct contact with the skin. It is used when the treatment area is relatively flat and smooth and larger than the transducer head (e.g., gluteal muscles).

Indirect coupling is placement of the transducer head at a distance to the skin. It

Figure 17.2 Ultrasound gel used as a coupling agent.

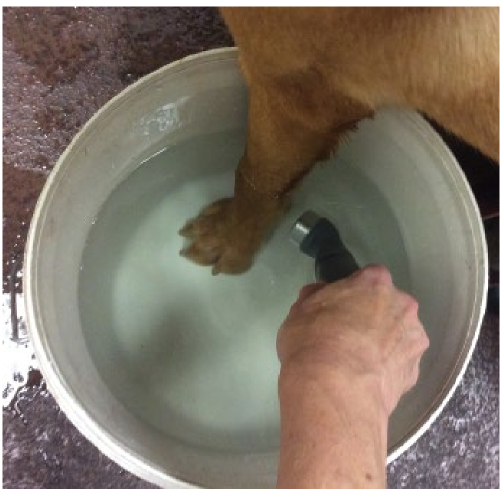

Figure 17.3 Treating a flexor tendon injury with indirect coupling. The patient's limb is immersed in warm water and the transducer head held 2 cm from the skin surface at the level of the lesion.

is used when the treatment area is irregularly shaped and smaller than the transducer head (e.g., digits). The most common form of indirect coupling is submersion, where the treatment area is submerged in a container of water and the transducer head is also submerged and held 1–3 cm from the skin's surface (Figure 17.3). Another type of indirect coupling is use of a coupling cushion, where a water balloon covered in coupling gel is placed between the skin and transducer head.

Role of the Veterinary Rehabilitation Technician

A technician that takes courses in veterinary rehabilitation will learn about therapeutic ultrasound and how and when to perform these treatments. If courses are not available to a technician, he or she can learn on the job from their veterinarian. The technician can perform these treatments without the supervision of the veterinarian in some states, however the treatment must always be prescribed by the veterinarian who employs the technician (legal supervisor). Prescription assigned to a technician by a non-supervising veterinarian is pushing the boundaries of state regulatory laws on the practice of veterinary medicine. After the treatment is completed, the technician should document that the treatment was performed and the treatment settings in the patient's medical records (Figure 17.4).

Figure 17.4 Maya, a 2-year-old Cockapoo having her Achilles tendon treated.

Clinical Applications

Conditions that benefit from non-thermal effects include acute soft tissue injuries, fractures, and wounds. Indications that benefit from thermal effects include osteoarthritis (except in acute exacerbation), muscle spasms, chronic soft tissue injuries, tendinopathies, muscle contracture, scar tissue and trigger points (Figure 17.5 and Box 17.1).

Dosage and Treatment Frequency

- Superficial conditions (target depth less than 2 cm) treated with pulsed ultrasound at 3.3 MHz should deliver between 0.25 and 0.9 W/cm^2 (Bockstahler *et al.*, 2004).
- Superficial conditions treated with continuous ultrasound at 3.3 MHz are recommended to be from 1 W/cm^2 to less than 1.5 W/cm^2. This is based on a research study by Levine and Millis in 2001, explained in the Relevant Scientific Research section (Levine *et al.*, 2001).
- Deep conditions are treated with either pulsed or continuous ultrasound at 1 MHz and 1–2 W/cm^2 (Steiss and Adams, 1999).
- Acute conditions are treated in short intervals (e.g., 3 times a week) for a short duration (e.g., 1–2 weeks) (Bockstahler *et al.*, 2004).
- Chronic conditions are treated in long intervals (e.g., 1–2 times a week) for a long duration (2–3 weeks or more if needed) (Bockstahler *et al.*, 2004).

Phonophoresis

Phonophoresis is the use of ultrasound to enhance the delivery and absorption of topically applied drugs such as analgesics

Figure 17.5 A feline patient undergoing ultrasound treatment of her neck muscles.

Box 17.1 Case study: Maya, 2-year-old spayed female Cockapoo

History

Maya was bitten on the right tarsus by her housemate during rough play.

Physical examination

On presentation, Maya was bright, alert and responsive. Her right tarsus was dropped and hyperextended, and there was a large swelling along distal calcaneal tendon, proximal to base of the gastrocnemius muscle.

Diagnosis

Partial right gastrocnemius and superficial digital flexor tendon avulsion

Treatment

Maya underwent reduction of the right gastrocnemius and superficial digital flexor tendon. A graft using a small portion of the tensor fascia lata was used to fill the 6 mm gap at the end of the gastrocnemius tendon. The deep digital flexor tendon was still intact therefore tension-relieving sutures were placed to decrease its length. Protein-enriched plasma was injected into the area of the tendons to help with healing. A bivalve cast was placed on the limb immediately postoperatively.

Two weeks postoperatively, a custom-made orthosis was used to provide stability in extension during the tendon healing phase, and a controlled and secure approach to tendon reloading. Adjustments were made to Maya's orthotic starting at 6 weeks postoperatively gradually allowing for 10 degrees increase in tarsal flexion up until 14 weeks postoperatively.

Maya received rehabilitation therapy consisting of therapeutic ultrasound to stimulate collagen formation with the results of allowing solid scar tissue formation and stability of the calcaneal tendon. The ultrasound settings used were 0.6 W/cm^2, 3 MHz, continuous wave, 7 minutes using a 5 cm^2 sound head. Low-level laser therapy was used to improve healing, decrease pain and inflammation, as well as alleviate muscle spasms of her paraspinals due to muscle compensation. Massage was performed to alleviate muscle compensation, and manual therapy was performed on all her limbs.

Therapeutic exercises were used to strengthen the muscles of the hamstring complex (semimembranosus, semitendinosus, and biceps femoris), quadriceps, and the gastrocnemius. Underwater treadmill was instituted to load the calcaneal tendon in a low-impact fashion using the buoyancy of the water.

Outcome

Maya completed 22 weeks of rehabilitation therapy starting 2 weeks postoperatively. Therapeutic ultrasound played an integral role in the healing of the gastrocnemius and superficial digital flexor tendon. At the end of the 22 weeks, she had functional mobility and was able to return to normal activities.

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and anti-inflammatories. Conditions in which phonophoresis might be of benefit include tendinitis, osteoarthritis, bursitis, sprains, and neuromas (Kleinkort and Wood, 1975).

Byl (1995) conducted a review of human literature and determined that to maximize the clinical effectiveness of phonophoresis the topical drug (both the drug and the carrying agent) should transmit ultrasound; the skin should be pretreated with ultrasound, heating, moistening, or shaving; the patient needs to be positioned to maximize circulation during treatment; a dressing that seals the area and prevents the escape of moisture should be applied after treatment; an intensity of 1.5 W/cm^2 should be used to capture both the thermal and nonthermal effects; and low-intensity (0.5 W/cm^2) should be used when treating open wounds or acute injuries. A study of phonophoresis using hydrocortisone was performed on the knees of greyhounds. Joint hydrocortisone levels obtained with phonophoresis were extremely low in comparison with those obtained with direct joint injection and the levels were not different to topical application (Muir *et al.*, 1990).

Clinical Reports

Mueller *et al.* (2009) published a case report of two cases with partial gastrocnemius muscle avulsion treated with pulsed therapeutic ultrasound. The first dog received 13 therapeutic ultrasound treatments (1 MHz, 25% pulsed waveform, 0.7–1 W/cm^2, 10 min) for 5 weeks. During the first 2 weeks, therapeutic ultrasound was performed 3 times per week, then decreased to 2 times per week. The second dog received 12 therapeutic ultrasound treatments (1 MHz, 25% duty cycle, 0.8 W/cm^2, 10 min), 3 times per week for 4 weeks. The outcome in both dogs was evaluated using ultrasonographic imaging and the measurement of ground reaction forces with a force plate. Both dogs showed an amelioration of the clinical signs within 1 month after commencement of the ultrasound therapy. The follow-up time for these cases was 1 year and 6 months, respectively. Both dogs were free of lameness and returned to their normal amount of exercise. Palpation of the fabella associated with the muscle injury did not produce any signs of pain. Ultrasonographic imaging did not detect any signs of hemorrhage or edema, although scarring of muscle fibers was present. The force-plate analyses revealed an improvement in lameness. These results suggest that therapeutic ultrasound could be a beneficial treatment modality for this kind of muscle injury (Mueller *et al.*, 2009).

In a clinical study of 124 cases of muscle injury in dogs treated with therapeutic ultrasound, patients were treated with 5 or 6 twice-weekly therapeutic ultrasound treatments. Settings were continuous wave, at a frequency appropriate for the depth of target tissue (most cases 3.3 Hz, a few larger dogs with deeper muscle injury were treated at 1 MHz) and at a power setting of 1.1–1.3 W/cm^2, depending on patient tolerance. A stretching program was instituted at ultrasound treatment number 3–5, depending on when muscle extensibility improved. Following ultrasound therapy, isometric strengthening exercises were instituted for 2–3 weeks (standing with one or two legs lifted), followed by sprint practice and slow return to full activity/sport. Lameness resolved in 91.9% of cases, but a small number recurred, needing long-term management with ultrasound therapy every 4–6 weeks. Lameness improved 1–2 grades for the remaining cases in which lameness did not resolve. The overall percentage of cases treated for muscle injury with therapeutic ultrasound with resolution of lameness and no recurrence/repeat injury was 85.8%. Of the sporting or working dogs, 93.9% fully returned to sport/work (Tomlinson, 2015).

Cautions and Contraindications

Thermal burns are a major concern and can be avoided by constantly moving the transducer head over the skin, and by using lowest effective intensity. The transducer head can overheat if held in the air while the unit is emitting ultrasound. Exert caution when using ultrasound over bony prominences, fracture sites, metal implants, artificial joints, irradiated skin, areas of decreased circulation, and areas of decreased pain and/or temperature sensation. Your supervising veterinarian will guide you and should write a specific prescription with treatment parameters, although this may need to change depending on patient response during treatment sessions. If the patient shows any sign of discomfort, immediately stop treatment. Palpate the treatment area for sensitivity and resume the same treatment settings only if it is confirmed that the patient was not reacting in pain and was just restless. A good way to determine this is to move the probe head as if treating, but with the ultrasound off. This will help to discern discomfort from ultrasound from discomfort from probe head contact or from restraint. If in doubt, stop and find the supervising veterinarian. The veterinarian will advise you about altering treatment settings (turning down the intensity for example) and when to cease treatment completely.

Therapeutic ultrasound is contraindicated over the heart, pacemakers, lower abdomen in a pregnant animal, eyes, testes, open epiphyseal plates, spinal cord, infection, ischemic tissue, hemorrhage, thrombosis, or malignancy.

Maintenance of Equipment

Because ultrasound therapy commonly produces little, if any, sensation, you must depend on your equipment to assure appropriate parameters are being delivered. To ensure that your equipment is in proper working order, it should be inspected regularly (e.g., annually) to assure proper calibration. Often, a local medical company will inspect and calibrate the machine; it does not have to be the machine's manufacturer. Keep the probe heads clean. The technician should do periodic (i.e., once every 2–4 weeks) maintenance of the equipment by testing the transducer heads. In order to test the transducer heads, place them in a cup of water and turn the machine on. If the transducer heads are working, a ripple effect will occur in the water. If a probe head is dropped for any reason, it will need to be checked and calibrated.

Relevant Scientific Research

In a study of five dogs with induced Achilles injury, tendons were cut and then sutured. A transcalcaneal screw was used to immobilize the tendon. One group received therapeutic ultrasound from the third day after surgery at 0.5 W/cm^2 for 10 minutes daily for 10 days. It appears that the ultrasound was used on continuous setting but it is not directly referred to by the authors. The healing of the Achilles tendon was monitored using clinical observations, ultrasonography, and histology. By day 40, the ultrasound appearance started to improve in the ultrasound-treated group compared to the untreated group. Gross observations suggested that the Achilles tendon in group 2 (ultrasound treated) showed comparatively fewer adhesions than in group 1 (control) animals. Histologically, the treated group had better healing across the tendon fibers, by day 120, the tendon tissue was close to normal. The authors concluded that ultrasound therapy at 0.5 W/cm^2 enhanced the Achilles tendon healing (Saini *et al.*, 2002).

Ten small (7–8 kg) dogs underwent bilateral ulna ostectomies resulting in a fracture gap. One side was treated with pulsed therapeutic ultrasound. Settings were reported as 1 MHz, 20% duty cycle with resultant energy delivered being 50 mW/cm^2. Treatment was performed for 15 minutes once a day for 6 days a week for 5 months. The low-intensity pulsed ultrasound enhanced new bone formation and decreased the incidence of nonunion (Yang and Park, 2001).

A study of the heating effect of continuous ultrasound was performed on the thigh muscles of 10 dogs in 2001 (Levine *et al.*, 2001). Treatment intensity was either 1.0 W/cm^2 or 1.5 W/cm^2. Temperature of the muscle was measured using thermistors at three different depths (1 cm, 2 cm, and 3 cm). Treatment time was 10 minutes. At the end of treatment, the temperature rise at an intensity of 1.0 W/cm^2 was 3 °C at a depth of 1 cm, 2.3 °C at 2 cm depth, and 1.6 °C at 3 cm depth. At 1.5 W/cm^2, temperature rose by 4.6 °C at 1 cm depth, 3.6 °C at 2 cm depth, and 2.4 °C at 3 cm depth. Tissue temperatures returned to baseline within 10 minutes or sooner after treatment. The authors stated that research suggests that a 3–4 °C increase in tissue temperature is effective in improving flexibility in both animals and humans. Based on these findings of temperature increase, intensity settings below 1.5 W/cm^2 continuous wave are recommended when treating dogs.

A study of the use of 3.3 MHz continuous wave ultrasound in horses evaluated the effects on temperature of digital flexor tendons and the epaxial muscles. Ten horses were evaluated, thermistors measured tissue temperatures. One tendon was treated for 10 minutes at an intensity of 1.0 W/cm and the other at 1.5 W/cm^2. Temperatures rose by 3.5 °C in the superficial digital flexor tendon and 2.5 °C in the deep flexor tendon treated with 1 W/cm^2 and by 5.2 °C in the superficial and 3.0 °C in the deep digital flexor tendon after 1.5 W/cm^2. The epaxial muscles were treated for 20 minutes, still using 3.3 MHz (a setting usually used for superficial tissue depths) at an intensity of 1.5 W/cm^2. Muscle temperature was measured at 1, 4, and 8 cm depths. Temperature rise was 1.3 °C at a depth of 1.0 cm, 0.7 °C at 4.0 cm, and 0.7 °C at 8 cm depth. The authors concluded that the digital flexor tendons are heated to a therapeutic temperature using a frequency of 3.3 MHz and intensity of 1.0 W/cm^2. The epaxial muscles are not heated to a therapeutic temperature using a frequency of 3.3 MHz and an intensity of 1.5 W/cm^2 (Montgomery *et al.*, 2013).

Freitas *et al.* (2010) evaluated the effects of therapeutic ultrasound on wound healing in rats. After 10 days of daily therapeutic ultrasound, results showed that therapeutic ultrasound improved wound healing, since the wound size had significantly reduced 5 and 10 days after starting treatment. Collagen levels were increased in the 0.6 W/cm^2 and 0.8 W/cm^2 treated groups. These authors concluded that therapeutic ultrasound has beneficial effects on the wound-healing process, probably by speeding up the inflammatory phase and inducing collagen synthesis (Freitas *et al.*, 2010).

Conclusions

Therapeutic ultrasound is a useful modality in veterinary physical rehabilitation. There is research showing the thermal effects in dogs and horses (at certain tissue depths), and confirming stimulation of bone and tendon healing in dogs. There are also some canine clinical reports showing successful treatment of muscle injury with this modality. More research and clinical data will come.

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18

Modalities Part 5: Shockwave Therapy

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Introduction

Extracorporeal shockwave therapy (ESWT), commonly known as shockwave therapy, uses high-energy sound waves called "pulses" or "shock waves" to stimulate and speed the body's own healing process. Shockwave therapy has been used in human medicine since the 1980s to provide a non-invasive treatment for urolithiasis (Ogden *et al.*, 2001). This technology then emerged as a modality to help with difficult-to-treat orthopedic and soft tissue injuries, including skin wounds. In the late 1990s this modality became of use in veterinary medicine in the field of equine medicine (Biggs, 2011). The high-energy sound wave creates mechanical stress at a cellular level leading to a release of healing growth factors in the body that reduce inflammation and swelling, increase blood flow, help bones heal, and enhance wound healing. Shockwave therapy can be used to help with a wide variety of conditions including tendon and ligament injuries (Danova and Muir, 2003; Hsu *et al.*, 2004; Hunter *et al.*, 2004; Venzin *et al.*, 2004), osteoarthritis (McIlwraith *et al.*, 2004; Francis *et al.*, 2004; Dahlberg *et al.*, 2005), wounds (Morgan *et al.*, 2009; Silveira *et al.*, 2010), and non-union fractures (Chen *et al.*, 2004). Shock wave is non-invasive so it can be used in these conditions in conjunction with other modalities to help your patients feel better and recover. It is important to note that not

Physical Rehabilitation for Veterinary Technicians and Nurses, First Edition.
Edited by Mary Ellen Goldberg and Julia E. Tomlinson.

all patients respond favorably to treatment with shockwave therapy. Although most studies show significant improvement after treatment, the results are not consistent (Brown *et al*., 2005). Lack of response may be attributed to ineffective treatment protocol or technique or to the patient's individual response to shockwave therapy.

When all things are considered, this modality is rather new in the realm of treating small animals. With more widespread use and further studies it is hoped that how this therapy is best used will be revealed and the protocols will be refined to provide more consistent results.

What is a "Shock Wave"

Shock waves are transient, short-term acoustic pulses with high peak pressure and a very short rise to peak pressure time on the order of magnitude of nanoseconds (one billionth of a second) (Mittermayr *et al*., 2012). To be considered a true "shock wave" the acoustic pulse must travel faster than 1500 m/s (the speed of sound).

There are three different ways to produce a true sound wave: electrohydraulic, electromagnetic, and piezoelectric. There are some therapy units that use radial pressure waves that travel at a much slower speed (~10 m/s) and therefore do not produce a true sound wave. The maximum energy is deposited on the skin and dissipates from there. Radial shockwave has been compared to a pneumatic "jack hammer" (Kirkby-Shaw, 2016). Research that shows positive effects of radial shockwave indicate that the dogs experienced a decrease in pain with a decrease in the degree of lameness (Mueller *et al*., 2007; Souza *et al*., 2016).

The electrohydraulic principle was used in the first generation of orthopedic shockwave machines. Electrohydraulic shock waves are high-energy acoustic waves generated by underwater explosion with high-voltage electrode spark discharge. The acoustic waves are then focused with an elliptical

Figure 18.1 An electrohydraulic shockwave machine.

reflector and targeted at the diseased area to produce the therapeutic effect. Electrohydraulic shock waves are characterized by large axial diameters of the focal volume and high total energy within that volume (Figures 18.1 and 18.2).

Shockwave generation through the electromagnetic technique involves the electric current passing through a coil to produce a strong magnetic field. A lens is used to focus the waves, with the focal therapeutic point being defined by the length of the focus lens. The amplitude of the focused waves increases by non-linearity when the acoustic wave propagates toward the focal point

The piezoelectric technique involves a large number (usually >1000) of piezo crystals mounted in a sphere, This receives a rapid electrical discharge that induces a pressure pulse in the surrounding water, steepening to a shock wave. The arrangements of the crystals cause self-focusing of the waves toward the target center, and lead to an extremely precise focusing and high energy within a defined focal volume.

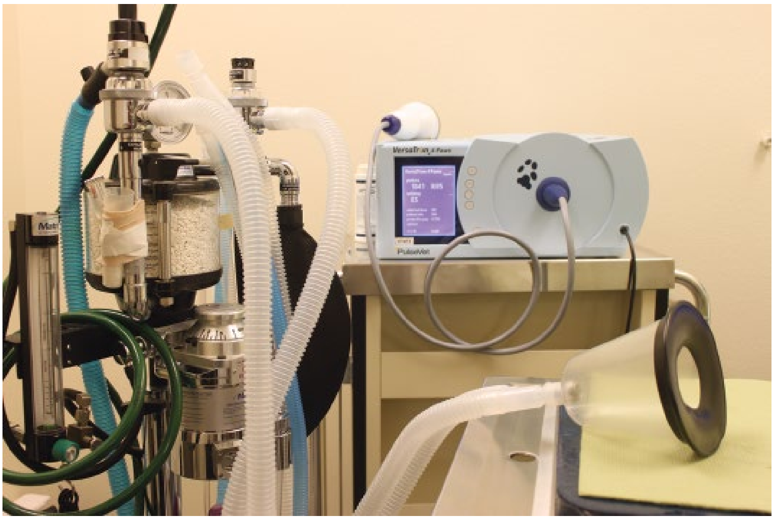

Figure 18.2 Versatron PulseVet™ next to anesthesia machine used for oxygen during sedation of patient.

When comparing different shockwave devices, the important parameters include pressure distribution, energy density, and the total energy at the second focal point in addition to the principle of shockwave generation of each device. For a complete therapy description, the energy flux density, in millijoules per millimeter squared (mJ/mm^2), the number of pulses, pulse repetition frequency, and the number and interval of retreatments are important parameters (Wang, 2012).

Mechanism of Action of Shock Waves

As the acoustic sound waves pass through tissue, energy is dissipated when it encounters denser tissue, such as bone, cartilage, tendon, and ligament. As energy is deposited in the tissues, the mechanical force translates to a biologic response that promotes healing. Currently the exact mechanism for the action for shockwave therapy in musculoskeletal disorders and the biologic effect on different anatomic structures (bone, cartilage, tendons, and ligaments) is not fully understood. It is believed that the mechanism of shock waves stimulates the early expression of angiogenesis-related growth factors including eNOS (endothelial nitric oxide synthase), VEGF (vessel endothelial growth factor), and PCNA (proliferating cell nuclear antigen) (Wang, 2012; Visco *et al.*, 2014). These factors then induce the ingrowth of neovascularization that improves blood supply and increases cell proliferation and eventual tissue regeneration to repair tendon or bone tissues. The rise of angiogenic markers was observed as early as 1 week after application and only lasted for approximately 8 weeks, whereas the neovascularization was first noted after 4 weeks and persisted for 12 weeks or longer, along with cell proliferation. These findings support the clinical observation that the effect of shockwave therapy appears to be dose-dependent and symptom improvement changes with time (Wang, 2012).

Other mechanisms might include: a release of cytokines, which can help a chronic condition revert to an acute state, thereby re-initiating the healing process with a reduction in inflammation and swelling; or release of BMP (bone morphogenic protein), which induces the formation of bone and cartilage.

Figure 18.3 An example of shockwave gel.

Patient Preparation

Acoustic sound wave energy dissipates as it passes from one medium to the next. Energy loss through attenuation is much larger through air than it is through water. Therefore, getting the probe head that delivers the sound wave as close to the skin as possible is recommended. The owner should be told that the area will need to have the hair short and a media gel applied to the skin (Figure 18.3). However, it is not necessary to shave the area.

Application of gel ensures good penetration of the sound waves as they pass from the probe head through the skin to the deeper tissues. It is along the boundaries between different media such as muscle and bone that the sound field experiences the biggest changes and emits the highest energy, and where the most biologic effects are expected.

After preparing the skin and applying an appropriate media gel, apply the probe head to the area being treated and use enough force to ensure good contact between the probe head and the skin (Table 18.1). Engage the shockwave generator and, as pulses are being formed, gently and slowly move the probe head over the area being treated, rotating and gliding the probe head as necessary to reach the intended target of treatment (Figure 18.4).

Table 18.1 Ranked couplant transmissivities relative to degassed water at 1.1 MHz and 3.4 MHz.

Couplant	Mean relative transmissivity at 1.1 MHz	Mean relative transmissivity at 3.4 MHz
KY	108%	99%
EMS	105%	99%
Aquasonic	104%	98%
JPM	100%	97%
Physiomed	98%	97%
SKF	95%	96%
Biofreeze	89%	88%

Source: Poltawski and Watson (2007). Reproduced with permission of Elsevier.

There is no consensus on how best to focus shock waves to manage a lesion. Some clinicians will use direct ultrasound guidance when treating a ligament or tendon injury and some will just estimate the area to be treated based on prior radiographic and sonographic diagnosis. Since focused shock waves are a locally concentrated therapy without regional or systemic effects, it is necessary to have a clear and concrete diagnosis to define the treatment area (McCarroll and McClure, 2002). A good working knowledge of the anatomical landmarks in the treatment area is essential to delivering the shock waves to the proper area and to avoid areas that the shock waves could damage or have less effect on. For arthritic joints, the best patient outcome seems to be obtained from concentrating on the attachments of the joint capsule surrounding the entire joint (Figures 18.5 and 18.6).

Typically, a loud “clicking” noise is created during the production of the shock wave and this noise may be startling to the patient. The intense stimulation from the energy produced

Figure 18.4 Treatment of canine elbow with probe head.

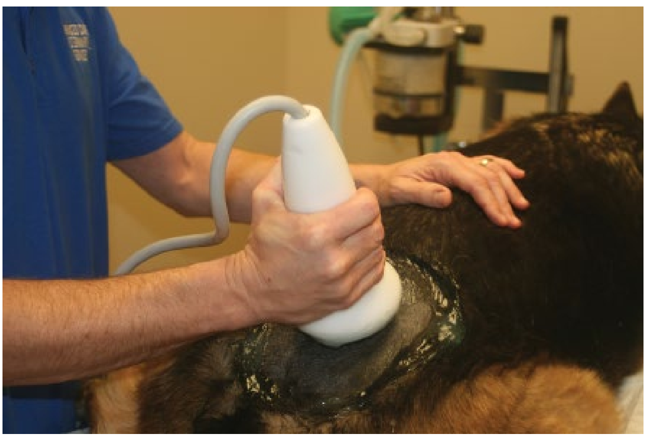

Figure 18.5 Treatment of canine pelvis with probe head.

by the shock wave may be perceived as uncomfortable to painful, depending on what condition is being treated and what shockwave generator is being employed. Therefore, mild to heavy sedation may often be used during the treatment sessions. Sedation can range from an alpha-2 (dexmedetomidine in small animals; detomidine or xylazine in equine) or a combination of an alpha-2 and opioid (butorphanol, hydromorphone, or morphine). This allows for comfort and stress relief to the patient and has the added benefit of safety for the veterinary staff.

What is the Common Treatment Protocol?

Treatment protocols have been established for many species, for a wide variety of indications and severity of disease. Because there are several different manufacturers and differences in the shockwave generators, it is imperative to follow the protocols set in place for each individual unit from the manufacturer. Most protocols call for two or three sessions of shockwave therapy at 2–3 week intervals. Be sure to enter the following information into

Figure 18.6 Treatment of feline elbow with probe head.

the patient's medical record for each session: type of shockwave generator used, energy flux density (mJ/mm^2), pulse repetition frequency, and the number of pulses.

The veterinary technician/nurse may be carrying out this treatment plan, but the rehabilitation veterinarian will be deciding upon the treatment range. The rehabilitation nurse will be instructed by the veterinarian on what and where to initiate shockwave therapy plus the settings used with the machine for each specific patient.

Pain Relief

Research in humans has shown that the analgesic effect of shockwave therapy appears to be more reliable in chronically affected patients rather than those acutely affected (Helbig *et al.*, 2001). Pain relief is quite common for 2–3 days after shockwave therapy (Buch *et al.*, 1998) and in an equine study analgesia was found to be similar to that induced by local or perineural analgesia, and the duration was 3 days (McClure *et al.*, 2004). However, the mechanism behind analgesia from shockwave therapy has not been elucidated. There has been some research in this area with no firm conclusions. Some research has even shown contradictory results: evidence that shows shockwave therapy has an influence on pain transmission by acting on substance P (Maier *et al.*, 2003; Hausdorf *et al.*, 2008) and calcitonin gene-related peptide (CGRP) (Takahashi *et al.*, 2003) expression in the dorsal root ganglion was refuted by the findings of Haake *et al.* (2002) who found no effect of shockwave therapy on substance P and CGRP. There is some evidence that shockwave therapy downregulates tumor necrosis factor-alpha (TNF-α) and interleukin-10 (IL-10) in osteoarthritic chondrocytes (Moretti *et al.*, 2008). The shock waves can create hyperstimulation of nociceptors resulting in intense stimulatory input into the periaqueductal gray area. This is the "gate control theory" whereby the nociceptive input is dampened before it reaches the brain; it has an abundance of supporting evidence (Rompe *et al.*, 1998; McClure and Weinberger, 2003; Bolt *et al.*, 2004; McClure *et al.*, 2005). Another theory is that cell damage is caused by the shock waves, thereby preventing normal membrane potentials required for the transmission of the nociceptive signals (Bolt *et al.*, 2004).

There are a wide variety of mechanisms that may allow for pain relief from shockwave therapy. Individual mechanisms may be affected by the type of shockwave generator used, the number of shock waves performed, energy level, frequency of application and the area being treated.

Adverse Events and Contraindications

When applied according to manufacturer specifications, the adverse events of shockwave therapy are minimal. The most commonly reported side-effect is bruising of the skin or soft tissue in the treatment area; small hematomas or swelling may also develop (Johannes *et al.*, 1994; Rompe *et al.*, 2001; McClure *et al.*, 2004; Dahlberg *et al.*, 2005; Speed, 2014). Shockwave therapy should never be directed over gas-filled cavities such as the lungs or the intestines (Ueberle, 1998). The acoustic impedance of air is distinctly lower than the acoustic impedance of soft tissue, thus virtually all the energy is reflected at the interface. The phase of the pressure is reversed and the maximum pressure at the interface may turn into rarefaction pressure, with up to twice the former maximal energy, which may result in considerable tissue damage at the interface. Intestinal perforation has been reported in a human after shockwave therapy where the colon was in the focal zone (Klug *et al.*, 2001). Large nerves and blood vessels also should not be within the focal zone. Arterial walls develop interstitial damage that can result in rupture, vasoconstriction, or increased permeability (Buch, 1998). Sites with neoplasia and infection have traditionally been avoided because of the potential to develop metastasis or septicemia from the primary site. The mechanical force of the shock wave can release the neoplastic cells and may cause an increase in the rate of metastasis (Oosterhof *et al.*, 1996; McClure *et al.*, 2004).

Indications

Osteoarthritis

Interest in the use of extracorporeal shock waves to treat orthopedic conditions was started by the observation that patients undergoing lithotripsy had an increase in pelvic bone density (Graff *et al.*, 1987). Shockwave therapy has been shown to modulate the osteoarthritic disease process in animal models by reducing eNOS, IL-10, and TNF-α levels and decreasing chondrocyte apoptosis, leading to a decrease in cartilage lesions (Moretti *et al.*, 2008; Zhao *et al.*, 2012; Kirkby, 2013). A significant improvement has been reported in ground reaction forces in dogs with elbow osteoarthritis after shockwave therapy as compared to the control group. These results suggest the effect of the therapy is what one would expect when treating this condition with a non-steroidal anti-inflammatory drug (Millis *et al.*, 2011). A study that included 14 dogs with stifle osteoarthritis showed an improvement in peak vertical force as well as range of motion that was not statistically different from the control group. However, in this study the dogs in the control group had a significant decrease in peak vertical force consistent with an increase in lameness (Dahlberg *et al.*, 2005). The evidence that shockwave therapy provides pain relief as part of a multimodal treatment plan is growing stronger each year and it can be a very valuable tool for managing arthritic pain (Mueller *et al.*, 2007; Souza *et al.*, 2016).

Tendon and Ligament Injuries

Tendinopathies typically result from overuse or repetitive activity and can cause significant pain and loss of partial or full function of a limb. These types of injuries are typically non-inflammatory in nature. The tendon's mechanical properties can break down due to an insufficient blood supply as well as degeneration of the tendon fibers. Learning from our equine colleagues, small animal practitioners have recently started to utilize shockwave therapy as part of the treatment strategy for their patients with tendon and ligament injuries. Evaluation of the bone–tendon interface in both the rabbit and dog model has demonstrated an increase in neovascularization, as confirmed by an increase in the angiogenic markers VEGF and eNOS and the formation of myofibroblasts that leads to healing after being treated with shockwave therapy (Wang CJ *et al.*, 2002, 2003). A study of 15 dogs that underwent shockwave therapy for issues of shoulder lameness after failure of conservative therapy found that 64% of the dogs remained improved or

Box 18.1 Indications for tendon and ligament healing

- Supraspinatus tendinopathy
- Biceps tendinopathy
- Medial shoulder instability
- Achilles tendinopathy (in addition to surgery if indicated)
- Patellar tendinopathy
- Iliopsoas strain

Source: Adapted from Kirkby (2013).

were without lameness at long-term follow-up, with a mean follow-up time of 844 days. These dogs presented with one of the following diagnosis: biceps tendinopathy; medial shoulder instability; supraspinatus tendinopathy; biceps and supraspinatus tendinopathy; medial shoulder instability and supraspinatus tendinopathy; or synovial osteochondroma (Becker *et al.*, 2015). Dogs that developed patellar desmitis after undergoing tibial plateau leveling osteotomy were evaluated after treatment with shockwave therapy. The thickness of the patellar ligament as measured at 6 and 8 weeks postoperatively was significantly different between treatment and control groups, demonstrating a positive effect of the shockwave treatment (Bosch *et al.*, 2007; Gallagher *et al.*, 2012) (Box 18.1).

Wound Healing

Large chronic wounds, especially those on lower limb extremities, can be quite challenging to treat. Research has demonstrated that the use of shockwave therapy may promote soft tissue healing because of neovascularization. Neovascularization will enhance the healing process and reduce pain associated with injury. One human study showed that 75% of patients had 100% epithelialization at a mean follow-up time of 44 days. The findings of this study suggest that unfocused low-energy shockwave therapy is a feasible modality for a variety of difficult-to-treat soft tissue wounds, particularly posttraumatic and postoperative wounds, decubitus ulcers, and burns (Schaden, 2007). Currently there are only experimental studies on equine wounds (Morgan *et al.*, 2009; Silveira *et al.*, 2010) and no published research in small animals. Further research is needed to determine the usefulness of this modality in veterinary medicine.

Bone Healing

Shockwave therapy is thought to increase bone healing through many different effects. Studies have implicated an induction of microfractures with the trabecular bone and the formation of medullary hematomas. The recruitment of bone marrow mesenchymal cells and the release of VEGF and transforming growth factor beta-1 (TGF-β1) may have a role in bone healing from shockwave therapy, and increases in BMP have been documented following shockwave therapy (Wang FS *et al.*, 2001, 2002, 2003). The growth factors in turn induce neovascularization and recruitment of osteogenic cells, leading to formation of bone (Chen *et al.*, 2004). In an equine study, no microfractures were induced and the mechanism was thought to be the result of bone marrow hypoxia, subperiosteal hemorrhage, increased regional blood flow, and activation of osteogenic factors such as BMP, direct cellular effects, and mechanical effects because of strain gradients (McClure *et al.*, 2004). In an experimental canine non-union radial fracture study, 4 out of 5 dogs had radiographic evidence of healing 6 weeks after a single shockwave session. At 9 weeks, there was narrowing of the fracture gap and increase in bridging callus. Complete radiographic confirmed union was present at 12 weeks. The one dog that had radiographic evidence of persistent non-union received a second shockwave session at 9 weeks after the first and had radiographic bony union in 12 weeks (Johannes *et al.*, 1994).

Wang (2012) demonstrated that shockwave therapy enhanced callus formation and induced cortical bone formation in acute fractures in dogs, and the effect of shockwave

Box 18.2 Indications for bone healing

- Delayed union and trophic non-union fractures
- In conjunction with surgical repair of highly comminuted fractures and/or anticipated slow fracture healing
- In conjunction with external coaptation for nonsurgical management of digital, metacarpal, or metatarsal fractures

Source: Adapted from Kirkby (2013).

therapy appeared to be time dependent. A study of electrohydraulic shock wave on the healing of stifle osteotomies (TPLO) showed that the radiographic healing score in the treated group of dogs was significantly better at 8 weeks than that in the untreated group (Kieves *et al.*, 2015). The current research is very supportive of the use shockwave therapy in treatment of delayed union and non-union fractures (Box 18.2).

Equine Use

Shockwave therapy entered veterinary medicine through its use in equine medicine, with the first recorded use in Germany in 1996 (McClure and Weinberger, 2003). The first use in the United States was in 1998 with horses with distal hock joint and navicular pain (McCarroll *et al.*, 1999; McCarroll and McClure, 2002). The research and use of shockwave therapy in equine medicine is vast when compared to that of small animal research. A number of studies have shown positive results in the treatment of: bone spavin, navicular syndrome, tendonitis, insertional desmopathy of the ligamentum nuchae, dorsal metacarpal disease, incomplete/stress fractures, proximal splint bone fractures, back pain due to "kissing spines" (Figure 18.7), and muscle pain.

In people and small animals skin petechiation at the treatment site has been reported, but it is not commonly seen in the horse (McClure and Weinberger, 2003). Despite the research not showing a clear mechanism, the analgesic effect is real and has caused some concern in the equine performance industry. According to the United States Equestrian Federation, most racing jurisdictions prohibit the use of shockwave therapy within 5–7 days preceding competition, and the Federation Equestre International prohibits shockwave therapy 5 days preceding competition (www.usef.org).

Clinical Applications

The following musculoskeletal conditions have been reported, either through clinical studies or anecdotally, to improve from the

Figure 18.7 Treating a horse for "kissing spine" disease.

Figure 18.8 Treatment of donkey lumbosacral disease.

use of shockwave therapy as the sole method of treatment or as an adjunctive treatment in a multimodal approach:

- chronic wounds
- lick granulomas
- delayed or non-union fractures
- non-surgical management of digital, metacarpal and metatarsal fractures
- osteoarthritis: hip, elbow, stifle, hock, shoulder
- supraspinatus tendinopathy
- glenohumeral ligament tears (medial shoulder instability)
- osteochondrosis of the shoulder
- triceps muscle tear
- fragmented coronoid process
- spinal column issues (intervertebral disc disease, lumbosacral disease, spondylosis deformans, cauda equine syndrome) (Figure 18.8)
- iliopsoas muscle injuries
- patellar tendonitis (especially following anterior cruciate ligament repair)
- Achilles tendinopathy
- calcaneal tendon tears.

Conclusion

Shockwave therapy is a unique modality that can be used to treat a wide range of conditions. Shock waves are high-energy acoustic waves generated under water from a wide variety of generators. The application of shockwave therapy for certain musculoskeletal disorders in veterinary medicine has been around since the late 1990s. This modality is non-invasive and has a low incident of adverse effects and complications. The current research demonstrates its effectiveness. However, further research is needed to understand the exact mechanism by which it works. Furthermore, due to a wide variety of soundwave generators, protocols need to be researched and developed for each condition treated, including energy level, number of shocks, frequency of pulses delivered, and number of sessions. Shockwave therapy should be considered as part of a multimodal approach to the conditions that have been discussed in this chapter.

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19

Therapeutic Exercises Part 1: Land Exercises

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CHAPTER MENU

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Introduction

Therapeutic exercises are a crucial component of any patient's rehabilitation program regardless of problem or diagnosis. Exercises need to be tailored to the individual patient considering the clinical diagnosis (a requirement before any exercise prescription), the age of the patient, physical condition, and resources available. Exercises can be used to improve a patient's range of motion (ROM), strength, weight bearing and gait pattern, their flexibility, balance and proprioception, and to decrease pain and improve healing. Exercises also improve strength, aerobic capacity (endurance) and performance, and help with weight loss. Therapeutic exercises as part of a home exercise program allow the clients to become involved in their pet's recovery and often strengthen the human–animal bond.

History of Therapeutic Exercise

Physical therapists have been utilizing therapeutic exercises in human medicine since the beginning of physical therapy in the early twentieth century. The aim is to improve function, performance, and disability (Saunders, 2007). Interest in applying these techniques to dogs started in the mid-1970s with the first publication of a book on canine rehabilitation techniques by Ann Downer MPT (Downer, 1978). Shortly

Physical Rehabilitation for Veterinary Technicians and Nurses, First Edition.
Edited by Mary Ellen Goldberg and Julia E. Tomlinson.

thereafter, national presentations about animal rehabilitation to such groups as the American College of Veterinary Surgeons (ACVS), American Physical Therapy Association (APTA), and the American Veterinary Medical Association (AVMA) led to more interest in canine rehabilitation. A growing interest in canine sports medicine resulted in the formation of the International Racing Greyhound Symposium in 1985. Rehabilitation was a frequent topic at this conference which later included all sporting dogs (McGonagle *et al.*, 2014). In the 1990s there was the long-awaited publication of two texts: *Care of the Racing Greyhound: A Guide for Trainers* (Craig *et al.*, 1994) and *Canine Sports Medicine and Surgery* (Bloomberg *et al.*, 1998). Today, physical rehabilitation is becoming common in small animal practices. Therapeutic exercises are used not only for recovery but also for wellness care and preventive medicine in the form of weight management and maintenance of muscle strength and conditioning, particularly for athletes and geriatric animals.

The Role of Exercise Physiology

Skeletal muscles perform motions guided by the nervous system. Skeletal muscle performance is dependent on muscle fiber type. Traditionally, muscles are classified as type I (oxidative or slow twitch) or type II (glycolytic or fast twitch) with subclassifications of these two types (Armstrong *et al.*, 1982). However, all muscles consist of a mix of different fiber types, in different ratios depending on the individual muscle and on training. Postural muscles (stabilizer muscles) such as the quadriceps femoris are capable of slow and sustained contraction and contain more type I fibers (about 50%) (Lieber *et al.*, 1989) than muscles like the gracilis, which contain more type II and are speed and power (mobilizing) muscles (Amann *et al.*, 1993). Type I muscle fibers have been thought of as the endurance muscle fibers (found to be more predominant in dogs which run long distances like sled dogs) and type II as the sprinting muscle fibers (for sprinting dogs such as Greyhounds) (Wakshlag *et al.*, 2004). However, when compared to humans, all dogs have a high oxidative capacity in all their muscles and are adapted for endurance activities (Wakshlag *et al.*, 2004). Certain breeds, for example Greyhounds, do have more fast twitch muscle fibers than others (Guy and Snow, 1981).

When muscles are immobilized, such as in casts or splints, muscle strength decreases rapidly, with as much as 50% of strength lost within the first week (Boyd *et al.*, 2009). With disuse, postural muscles that contain a predominance of type I fibers atrophy more than the mobilizing muscles containing type II fibers. With geriatric sarcopenia (the age-related change in muscles), it appears that in dogs the epaxial muscles atrophy early in the process. A study of live dogs compared epaxial, quadriceps, and temporalis muscle size in aged dogs, and in size and body condition score matched young dogs. The results showed that epaxial muscles were smaller in aged dogs but quadriceps and temporal muscles did not differ significantly (Hutchinson *et al.*, 2012). Fiber type lost, however, appears preferential for the large type II (power and strength) fibers (Deschenes *et al.*, 2013), as shown in analysis of cadaver canine biceps femoris muscles (Pagano *et al.*, 2015). This is an important consideration in designing an exercise program for an athlete with muscle loss due to injury, versus a geriatric patient with age-related sarcopenia (Appell, 1990).

Muscle contractions can be described as having two variables: force and length. The force is either tension or load. Load is the force exerted on the muscle by an object and muscle tension is the force the muscle exerts on an object. Isometric contractions occur when muscle tension changes with no change in muscle length. This is a static exercise, for example lifting a front leg so that more weight/load is on the muscles of the contralateral front limb and the rear limbs; these limb muscles are undergoing isometric contractions to support more body weight. Tension bands applied to the standing patient rely on the instinct to lean into pressure. As a patient maintains the same body position under an increased load (whether push or pull), the muscle work has increased without a change in muscle length.

Isotonic contractions occur when the muscle tension remains the same but the muscle length changes. Isotonic contractions occur as either concentric or eccentric contraction. Concentric contraction occurs when tensions in the muscle increases along with shortening. An example of this would be a human weight lifter performing a biceps curl. Eccentric contraction occurs when the muscle contracts but lengthens because the tension generated in the muscle is insufficient to overcome the load pulling down on the muscle. Think of the weight lifter slowly releasing the biceps curl and extending their elbow to put down the weight. The eccentric contraction is controlling the movement – it is the natural braking force that occurs during motion (Gillette and Duke, 2014). Eccentric contractions can predispose to injury in untrained individuals (Whitehead *et al.*, 2003). Resistive exercise of all forms leads to the preferential hypertrophy of type II fibers and eccentric contractions render type II fibers more susceptible to damage when compared to type I fibers in humans, rabbits, and rodents (Quindry *et al.*, 2011). In an exercise program, generally, concentric exercises are performed first to help accustom the muscle to movement. Eccentric exercises are added later as these have the potential to cause damage to the muscle and delayed-onset muscle soreness, but they help develop greater strength. A balanced program between concentric and eccentric muscle contractions is desired.

Principles of an Exercise Program

When designing an exercise program for a patient the therapist must consider any pathology affecting the cardiovascular, neuromuscular, skeletal, pulmonary, endocrine/metabolic, or integumentary system. These systems can affect muscular performance; for example a dog with laryngeal paralysis will have some respiratory compromise, and a patient with a repaired ligament will have some reduced joint stability early in the healing process. It is also important to evaluate the patient's psychological state and willingness to perform any exercise as well as the experience of the client or handler. A home exercise program may be very different for an agility dog with an experienced trainer versus a geriatric companion dog with an aging client.

After evaluating the patient, identifying problems and the amount of tissue healing that has already taken place (tissue integrity), the therapist must set both long- and short-term goals for the patient. It is important that the client's goals are in line with those of therapist. Each part of the program – proprioception, strength, endurance, and power (speed plus strength) – should have specific goals and these will vary with the patient and with the injury. Exercises should also include relaxation time between bouts, and flexibility exercises specifically tailored to the demands made on the body. It is important for therapists to know as much as we can about the effects of the exercises we use. Although not a lot of information is known about all the exercises we use, where information exists we should apply it. For example, cross walking (serpentine) on a hill results in a temporary increase in loading of the downside hindlimb, transferring forces off the upside hindlimb, but dogs adapt quickly and compensate – more weight is shifted to the downside forelimb and shorter steps are taken (Strasser *et al.*, 2014). Using a serpentine pattern to start to introduce hills, or to start to increase the workload on a recovering hindlimb may not have the desired effect of significantly increasing the load on that limb beyond the first few steps therefore a different exercise may be more appropriate.

Once the goals are set and the program is designed, it is important for the therapist to evaluate the patient at each visit. Asking and documenting questions about soreness and level of home activity after the last session is crucial. The rehabilitation technician/nurse is instrumental in acquiring this information. This allows the therapist to adjust the program as needed so the patient continues to progress through the rehabilitation process. Therapists should ensure proper, efficient mechanics in motion for their patients. Faulty patient body mechanics must be corrected wherever possible and good fundamentals ingrained. Use a hand or an assistive device to guide as needed. Watch for decline of movement patterns (in gaiting and during transitions) as signs of fatigue. Do not continue an exercise once body mechanics degenerate from the best possible for that patient at that time (even if that best had initially required some assistance). Rest for a period (usually a few minutes), then attempt to resume the exercise. If the exercise cannot be performed to the level it was earlier in the session, then end the session. Ask for an easy motion to end on a good note.

At no time should the patient be harmed or uncomfortable. Chronic overexertion and fatigue can increase susceptibility to injury. According to the International Associations of Athletics Federations online medical manual (IAAF, 2016) muscular overexertion may present as muscle soreness, muscle stiffness, and muscle spasm. Adaptation to the demands of strengthening exercises occur gradually, over long periods of time. Efforts to accelerate the process may lead to injury. Conversely, an inadequate training load will not provide an adequate stimulus, and a strengthening of tissues will not occur.

Certain activity patterns are especially likely to cause overtraining (IAAF, 2016). These include a sudden increase in training volume and/or intensity without a gradual build-up and use of a single, monotonous training format which fatigues one muscle group or energy system. Early warning signs of overtraining include a longer recovery time needed between exercise bouts (patient is tired and stiff for longer than 2 hours after exercise). In this case the exercise may be too intense or the patient may need re-evaluation. Icing post exercise and prophylactic pain control will be needed, particularly for postoperative patients. In weak, ataxic or non-ambulatory patients, assistive devices must be used for exercise and this can include booties to protect feet from scuffing the floor or harnesses or slings to assist in ambulation.

According to McCauley and Van Dyke (2013), there are five variable parameters in any exercise program:

1) Frequency of work done (multiple times per day, daily or weekly)
2) Speed/intensity
3) Duration of work (time or number of reps)
4) Environment (terrain, footing, substrate)
5) Impact (low, high or no impact).

As the patient heals, the frequency, intensity, and duration of the exercise is increased to further challenge the patient and to strengthen the muscles. Remember that tendons, bones, and articular cartilage also remodel and "strengthen" with a correctly targeted exercise program. It is usually safe to increase the activity by 10–15% each week if the patient does not experience an increase in pain or a loss of function (Millis *et al.*, 2014).

Canine Rehabilitation Equipment

Physioballs/Peanuts/BOSU® Balls

Exercise balls come in many shapes and sizes and have many different uses. Peanut balls look exactly like a peanut shell, with an indent in the middle providing two separate points of ground contact for added stability over an oval ball. A physio roll is like a peanut ball but lacks the middle indent. Egg-shaped balls have less stability, which makes exercises on these more challenging. Round balls are the most challenging as they allow movement in all directions. A BOSU® ball is flat on one side and has a half ball attached, allowing the patient to balance on the half ball side or when flipped over to balance on the flat side (Figure 19.1).

In general, most ball work will start with an underinflated peanut for the most stability. The therapist should stabilize the peanut and allow for only small motions. Make sure the patient has good posture, avoid strengthening one muscle group predominantly over another (e.g., spinal flexors over extensors, resulting in kyphosis), always follow with a stretch to reverse the motion. As the patient progresses, more air is added for an additional challenge; later, more challenging ball shapes, along with more challenging postures, are used. When introducing a patient to the ball, make it a positive experience by using treats (see Chapter 9). When using balls for exercises, remember that time can be lengthened for sustained muscle contraction and transitions on the ball can be used for improving both strength and control further. Be cognizant of the type of muscle contraction you are asking for, and hence the muscle fiber type you are targeting. An old dog with weak spinal stabilizing muscles will benefit more from standing still in good posture on a challenging surface, rather than attempting transitions.

- Physioballs can help build strength while also increasing balance and proprioception.
- Peanut balls and BOSU balls are used for beginners as they allow more stability due to increased ground contact. BOSU balls are particularly good for core strengthening.

Cavalettis

Cavaletti poles or ground poles have been used extensively for training horses. In the small animal patient, these poles are used to train gait, improve proprioception, and strengthen the forelimb and hindlimb flexor muscles (Figure 19.2). They are also used to improve active ROM – specifically they increase flexion and extension of stifle and flexion of the carpus and elbow along with flexion of the tarsus (Holler *et al.*, 2010). Cavalettis are placed in a series, and are adjustable in height. They are placed

Figure 19.1 The patient has all four paws on the BOSU ball to work on balance. Posture is corrected as needed.

Figure 19.2 Cavalettis are great for strengthening the forelimbs and hindlimbs and as a proprioception exercise.

low or on the ground when the patient begins the exercise, or has significant muscle weakness. The height can be adjusted as the patient progresses through rehabilitation. They can also be placed in a circular pattern for jumping or as a series of "bounce" jumps for more advanced patients or athletes (Millis *et al.*, 2014). Spacing is patient-dependent; it depends roughly on height and body length, but most importantly on stride length.

Cavalettis are very easy to make using pylons (cones) with holes and PVC pipe or crushed aluminum cans and 2 × 4-inch planks. For home use clients can use anything from PVC pipe to broomsticks or even pool noodles, depending on the size of the dog. The cavalettis are spaced so that only one paw at a time is placed between the poles in gaiting over them, thus challenging proprioception as the dog avoids touching the pole. The height of the pole determines how much flexion occurs; more flexor strengthening occurs with higher poles.

- Cavalettis are used for improved proprioception and strengthening. They are a good exercise for dogs ranging from weak geriatric to athletes. They are easy for the client to make at home.

Weave Poles and Cones

These poles and cones are used for circling, walking in a figure eight, and weaving (in a serpentine). Weaving in and out of cones creates lateral flexion of the spine, aims to strengthen the adductor and abductor muscles and to improve balance and proprioception. Six to eight traffic cones can be used to make up an obstacle course for the dog to weave in and out of. Alternatively, multiple objects such as bowling pins, water bottles, or a line of trees can be used if they are lined up evenly spaced so the dog can weave in and out of the objects. Vertical weave pole agility sets can be used. An alternate is traffic cones with a pole at the top. The distance between the poles needs to be adjusted so that sufficient lateral bending occurs.

- Weaves can be used to visualize any discrepancy in the patient's gait. The objects can be anything that is lined up in a straight line. Weaves are used to help build core muscles and improve proprioception.

Planks/Blocks/Stairs

Planks are 2 × 8 inch (5 × 20 cm) or 2 × 10 inch (5 × 25 cm) pieces of wood that are 8–10 feet (2.5–3.0 m) in length. The planks are initially placed on the floor and the dog walks along it while maintaining balance. The plank is then raised up onto cinder blocks. The dog is further challenged by placing blocks as obstacles on the plank.

Blocks are smaller, thicker pieces of wood with a 4 × 6 inch (10 × 15 cm) non-slip area. Grip tape can be added (skateboard product). They are made in sets of 2, 4, and 6 inch (5, 10, and 15 cm) heights. The dog stands with one foot on each block or any combination of diagonals or front or back paws. Strengthening of the stabilizer muscles of the trunk is emphasized by this exercise.

Stairs can be made of any material and many different types of stairs exist in a home or clinical setting. Climbing stairs is good for proprioceptive training, core muscle strengthening, improved hindlimb weight bearing, and improving ROM of pelvic limb joints (improved extension of hip and hock and improved/increased flexion of stifle and hock) (Durant *et al.*, 2011). Descending stairs is also good for balance and proprioception and should increase forelimb weight bearing.

- Planks, blocks, and stairs strengthen and target proprioception.

Balance Discs or Boards

Balance discs are rounded on both sides, not as curved as a ball, with a side rim of about an inch (2.5 cm) high. Each surface has a different texture. One is smooth, the other is textured and can so provide more grip.

Figure 19.3 The rocker board pictured is placed against the wall. The dog is lured to place a foot on the board.

Balance discs create an uneven surface that helps to improve balance and strength, in both the digital flexors and in the stabilizer muscles that are activated by the uneven stance the balance disc creates (Figure 19.3). Balance discs are easier to stand on than are balls and can be used alongside other stabilization exercises.

Balance boards are created by laying a piece of plywood over a pillow, water bottle, or ball, thus creating an unstable surface. These can be made by gluing a half tennis ball to the bottom of a piece of plywood that has been covered in indoor/outdoor carpeting, making a non-slip surface.

Limb Weights

Limb weights are small weights that can easily be secured to the limbs. Normally they have a Velcro strap attached to them. With some creativity, limb weights can be manufactured for patients of different sizes. For some dogs, we have used stainless steel washers of various sizes that are readily available in the hardware store. These are wrapped in vet wrap and placed on the limb. Curtain weights also work. The advantage of these small structures is that you can use numerous ones of varying weights to slowly increase the weight on the limb. Limb weights are for more advanced rehabilitation and they promote natural weight shifting on the diagonal. Commercially available weights are available (e.g., from Canine Icer™).

Elastic Resistant Bands

These (e.g., TheraBands) are 6-inch-wide (15 cm) latex resistance bands that are color coded for different resistance levels, which are determined by the thickness of the material. They can be used in both beginner and advanced therapy and can also be used as a means of pulling a paralyzed leg forward, mimicking regular gait. They can also be used to increase tension and resistance to facilitate muscle development.

Trampoline (Mini)

This is a very useful tool for training proprioception and balance in small dogs and cats. The trampoline causes an uneven surface area that in turn causes multiple muscle firings. It is an excellent tool to increase core strength in small dogs recovering from back surgery and to build up core strength to help prevent back problems in this same group of dogs.

Land Treadmills

Land treadmills provide a great workout for dogs and cats, improving limb strength and increasing cardiovascular endurance. Neurological patients can benefit from treadmills for gait patterning during recovery from paralysis. A board placed across the front of the treadmill allows the front feet to be elevated off the belt while

Box 19.1 Treadmill safety

- Always use a leash and harness with the patient. Collars can be dangerous
- Never tie the patient to the treadmill or leave the patient unattended. Stand next to the dog throughout the entire workout
- Do not face the treadmill into a wall – the dog will resist walking "into" the wall
- Lead the patient onto the treadmill using an incentive such as a treat or toy, and then the treadmill can be slowly turned on
- If your treadmill does not have safety walls you may need assistance to keep the pet's attention looking and walking forward. Place one side against the wall so the patient does not fall off the side
- Short intervals are important until the patient gets accustom to the treadmill. Go slow and let the patient get acclimatized to the routine
- Always monitor the amount of panting, the pet's gait, body language, and signs of fatigue (excess or rapid panting, glazed eyes, change to gait (wobbling, staggering) and drooping tongue) the entire time
- Allow rest periods where stretching, massage, and ROM can be performed
- A water dish can be offered during intermission time as well
- Remember that each session should be positive and time on the treadmill should be dictated by the patient's condition and response

hindlimb gait retraining is concentrated on. Small patients can be placed in their cart on a treadmill. Most treadmills can be inclined (or in some cases down to a decline) to work on improving strength or reducing the force placed on front or hind legs. It has been reported that dogs walking on a treadmill with a 5% incline had increased hamstring activity, but that gluteals and quadriceps were not affected (Lauer *et al.*, 2009). So an incline should improve hamstring strength.

The speed and length of time can be varied to build up endurance and for conditioning. Canine athletes can be conditioned on treadmills with various inclines and various speeds for intervals of 20–30 minutes at a fast trot. Most medium to large dogs will comfortably walk at a rate of approximately 2 mph (3.2 km/h) whereas neurological patients must be started at 0.1–0.5 mph (0.2–0.8 km/h). It is important that the therapist evaluate each dog on the treadmill to determine optimal speed and effort while maintaining a normal gait (Box 19.1). What works for a Golden Retriever may not work for a Border Collie!

Dogs that are leash walked can be easily trained to walk on a treadmill (Box 19.2). A variety of treadmills can be used for exercising dogs. Specific dog treadmills such as Jog a Dog (inclined treadmills) or Dog Pacers can be purchased, or a human treadmill can be used (Figure 19.4). If a human treadmill is used, the belt length needs to be long enough to accommodate the dog's stride. Many medium and large dogs have too long a stride length for the traditional treadmill, especially at trot. Remember that during normal gaiting on land, speed varies. Also, a treadmill has the potential to gait retrain a patient into an undesirable pattern. Land treadmills can be used for feline patients if the cat is taught slowly (Figure 19.5).

Desirable features for a land treadmill include (McCauley and Van Dyke, 2013):

- belt length long enough to accommodate expected size of dog (minimum 6 feet/2 m)
- incline/decline capacity
- ability to go in reverse direction (hence decline?)
- one button push-start/stop or turn of a knob
- exact speed visible (this allows consistent exercise intervals)
- starting speed of 0.1–0.2 mph (0.2–0.3 km/h)
- side rails.